BOOK 8
ELECTRICITY

9 Electricity

9.1 Electric current

Candidates should be able to:

1 understand that an electric current is a flow of charge carriers

2 understand that the charge on charge carriers is quantised

3 recall and use $Q = It$

4 use, for a current-carrying conductor, the expression $I = Anvq$, where n is the number density of charge carriers

9.2 Potential difference and power

Candidates should be able to:

1 define the potential difference across a component as the energy transferred per unit charge

2 recall and use $V = W / Q$

3 recall and use $P = VI$, $P = I^2R$ and $P = V^2 / R$

9.3 Resistance and resistivity

Candidates should be able to:

1 define resistance

2 recall and use $V = IR$

3 sketch the I–V characteristics of a metallic conductor at constant temperature, a semiconductor diode and a filament lamp

4 explain that the resistance of a filament lamp increases as current increases because its temperature increases

5 state Ohm's law

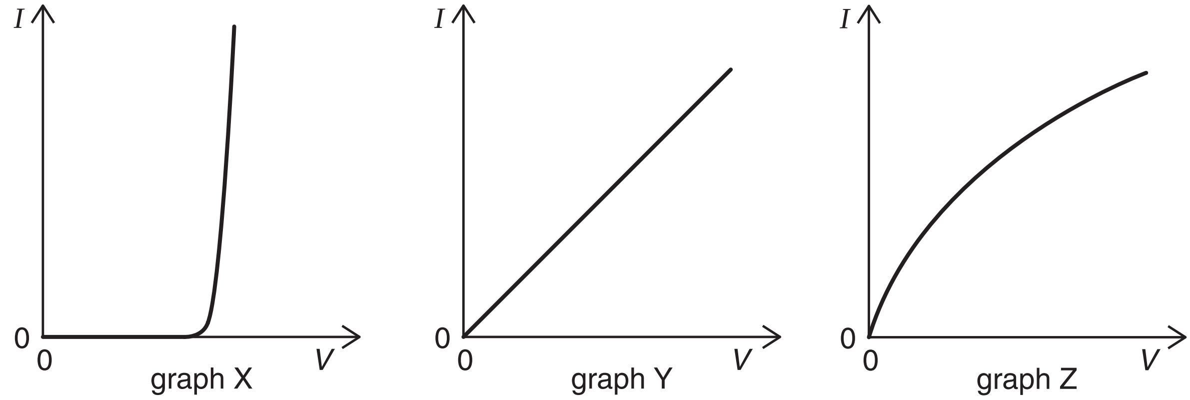
6 recall and use $R = \rho L / A$

7 understand that the resistance of a light-dependent resistor (LDR) decreases as the light intensity increases

8 understand that the resistance of a thermistor decreases as the temperature increases (it will be assumed that thermistors have a negative temperature coefficient)

1 **9702/1/M/J/02/Q30**

The graphs show the variation with potential difference V of the current I for three circuit elements.



The three circuit elements are a metal wire at constant temperature, a semiconductor diode and a filament lamp.

Which row of the table correctly identifies these graphs?

	metal wire at constant temperature	semiconductor diode	filament lamp
A	X	Z	Y
B	Y	X	Z
C	Y	Z	X
D	Z	X	Y

2 **9702/1/O/N/02/Q30**

Which equation is used to define resistance?

A power = $(\text{current})^2 \times$ resistance

B resistivity = resistance $\times$ area $\div$ length

C potential difference = current $\times$ resistance

D energy = $(\text{current})^2 \times$ resistance $\times$ time

3 **9702/01/M/J/05/Q32***

A copper wire of cross-sectional area 2.0 mm^2 carries a current of 10 A.

How many electrons pass through a given cross-section of the wire in one second?

A 1.0×10^{1} **B** 5.0×10^{6} **C** 6.3×10^{19} **D** 3.1×10^{25}

4 **9702/01/M/J/05/Q33**

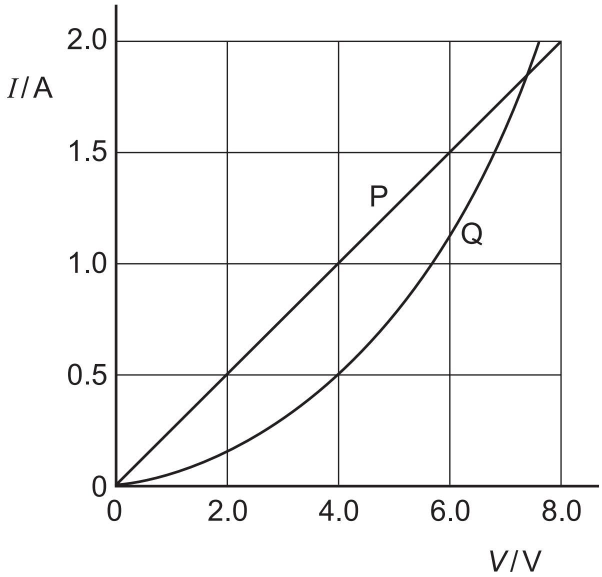
A cylindrical piece of a soft, electrically-conducting material has resistance R. It is rolled out so that its length is doubled but its volume stays constant.

What is its new resistance?

A $\frac{R}{2}$ **B** R **C** $2R$ **D** $4R$

5 9702/01/M/J/05/Q34

The I-V characteristics of two electrical components P and Q are shown below.



Which statement is correct?

A P is a resistor and Q is a filament lamp.

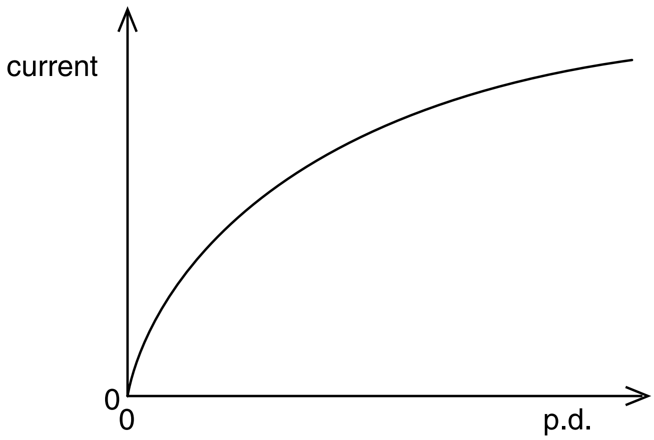
B The resistance of Q increases as the current in it increases.

C At 1.9 A the resistance of Q is approximately half that of P.

D At 0.5 A the power dissipated in Q is double that in P.

6 9702/1/O/N/02/Q31

The graph shows how the current through a lamp filament varies with the potential difference across it.



Which statement explains the shape of this graph?

A As the filament temperature rises, electrons can pass more easily through the filament.

B It takes time for the filament to reach its working temperature.

C The power output of the filament is proportional to the square of the current through it.

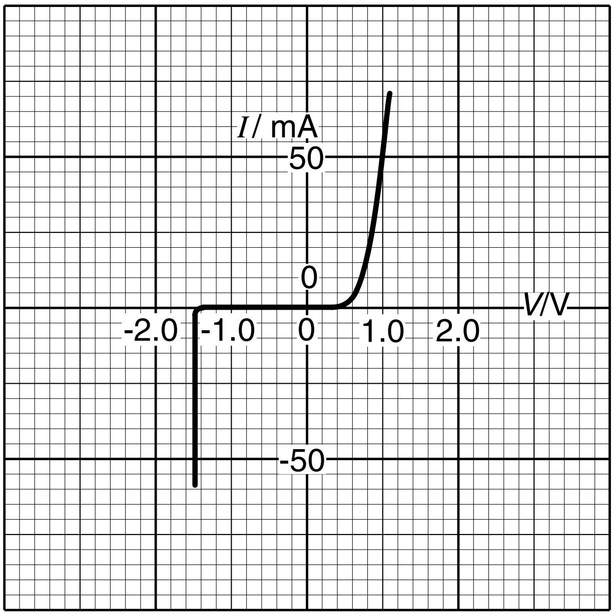
D The resistance of the filament increases with a rise in temperature.

7 **9702/1/O/N/02/Q32**

The variation with potential difference V of the current I in a semiconductor diode is shown below.

What is the resistance of the diode for applied potential differences of +1.0 V and –1.0 V?

	resistance	
	at +1.0 V	at –1.0 V
A	20 Ω	infinite
B	20 Ω	zero
C	0.05 Ω	infinite
D	0.05 Ω	zero

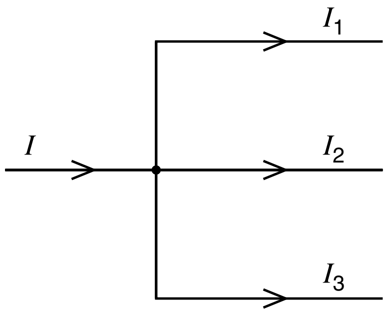


8 **9702/1/O/N/02/Q33**

At a circuit junction, a current I divides into currents I_1, I_2 and I_3.

These currents are related by the equation $I = I_1 + I_2 + I_3$.

Which law does this statement illustrate and on what principle is the law based?



A Kirchhoff's first law based on conservation of charge

B Kirchhoff's first law based on conservation of energy

C Kirchhoff's second law based on conservation of charge

D Kirchhoff's second law based on conservation of energy

9 **9702/12/O/N/14/Q35**

The combined resistance R_T of two resistors of resistances R_1 and R_2 connected in parallel is given by the formula

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

Which statement is used in the derivation of this formula?

A The currents through the two resistors are equal.

B The potential difference across each resistor is the same.

C The supply current is split between the two resistors in the same ratio as the ratio of their resistances.

D The total power dissipated is the sum of the powers dissipated in the two resistors separately.

10 **9702/01/M/J/03/Q29**

What physical quantity would result from a calculation in which a potential difference is multiplied by an electric charge?

A electric current **B** electric energy **C** electric field strength **D** electric power

11 **9702/01/M/J/03/Q30**

The current in a component is reduced uniformly from 100 mA to 20 mA over a period of 8.0 s.

What is the charge that flows during this time?

A 160 mC **B** 320 mC **C** 480 mC **D** 640 mC

12 **9702/01/O/N/03/Q30**

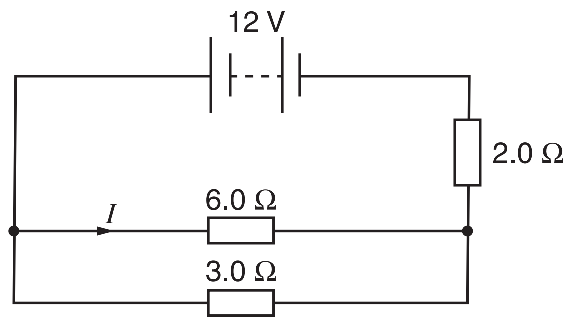
A wire carries a current of 2.0 amperes for 1.0 hour.

How many electrons pass a point in the wire in this time?

A 1.2×10^{-15} **B** 7.2×10^{3} **C** 1.3×10^{19} **D** 4.5×10^{22}

13 **9702/01/O/N/03/Q31**

The diagram shows a circuit in which the battery has negligible internal resistance.



What is the value of the current I?

A 1.0 A **B** 1.6 A **C** 2.0 A **D** 3.0 A

14 **9702/01/M/J/04/Q31**

Two wires made of the same material and of the same length are connected in parallel to the same voltage supply. Wire P has a diameter of 2 mm. Wire Q has a diameter of 1 mm.

What is the ratio $\dfrac{\text{current in P}}{\text{current in Q}}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** 2 **D** 4

15 **9702/01/M/J/04/Q32**

What is an equivalent unit to 1 volt?

A $1\,\mathrm{J\,A^{-1}}$ **B** $1\,\mathrm{J\,C^{-1}}$ **C** $1\,\mathrm{W\,C^{-1}}$ **D** $1\,\mathrm{W\,s^{-1}}$

16 **9702/01/M/J/04/Q33**

The terminal voltage of a battery is observed to fall when the battery supplies a current to an external resistor.

What quantities are needed to calculate the fall in voltage?

A the battery's e.m.f. and its internal resistance

B the battery's e.m.f. and the current

C the current and the battery's internal resistance

D the current and the external resistance

17 **9702/01/M/J/04/Q34**

The potential difference between point X and point Y is 20V. The time taken for charge carriers to move from X to Y is 15s, and, in this time, the energy of the charge carriers changes by 12J.

What is the current between X and Y?

A 0.040A **B** 0.11A **C** 9.0A **D** 25A

18 **9702/01/O/N/04/Q31**

Which of the following describes the electric potential difference between two points in a wire that carries a current?

A the force required to move a unit positive charge between the points

B the ratio of the energy dissipated between the points to the current

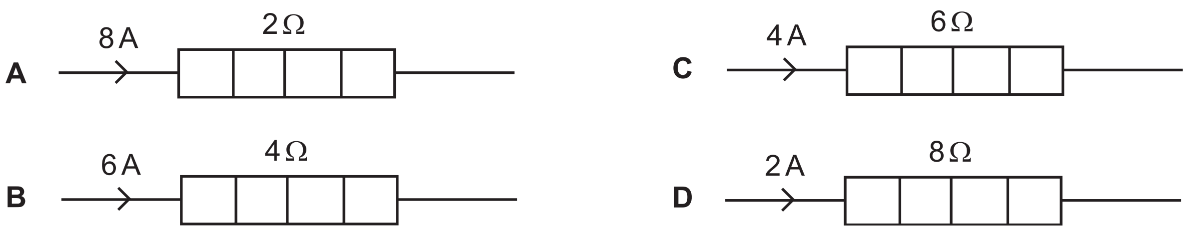
C the ratio of the power dissipated between the points to the current

D the ratio of the power dissipated between the points to the charge moved

19 **9702/01/O/N/04/Q32**

The diagram shows four heaters and the current in each.

Which heater has the greatest power dissipation?



20 **9702/01/O/N/04/Q33**

When a potential difference *V* is applied between the ends of a wire of diameter *d* and length l, the current in the wire is I.

What is the current when a potential difference of 2*V* is applied between the ends of a wire of the same material of diameter 2*d* and the length $2l$? Assume that the temperature of the wire remains constant.

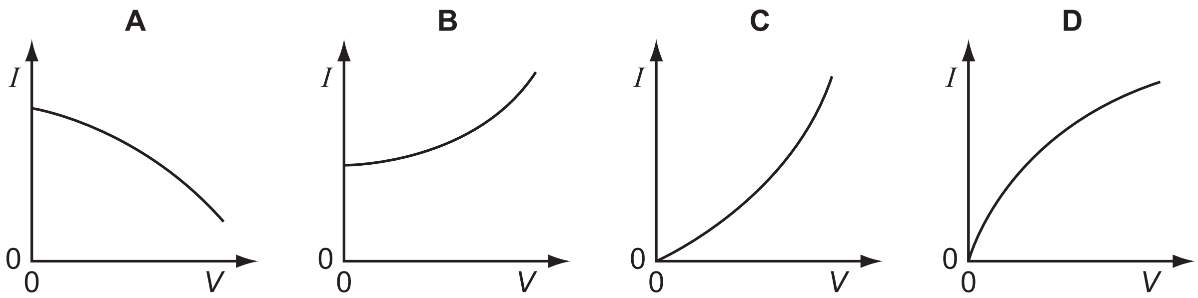
A I **B** $2I$ **C** $4I$ **D** $8I$

21 **9702/01/O/N/04/Q34**

The resistance of a thermistor decreases significantly as its temperature increases.

The thermistor is kept in air. The air is at room temperature.

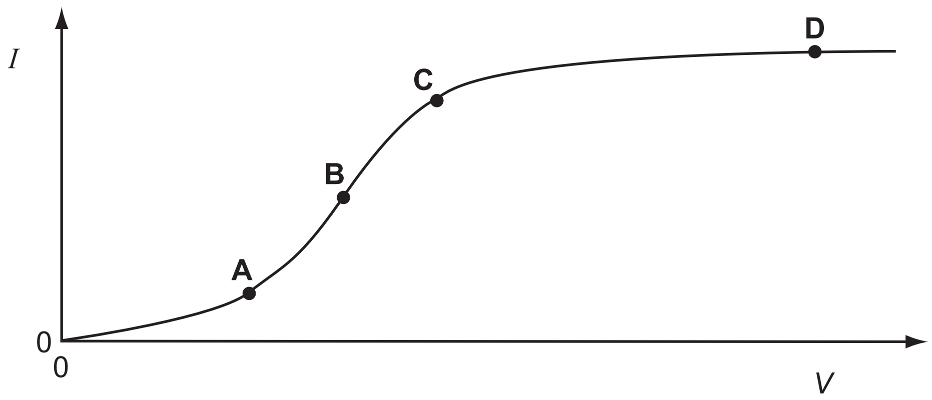
Which graph best represents the way in which the current I in the thermistor depends upon the potential difference *V* across it?



22 **9702/01/O/N/05/Q34**

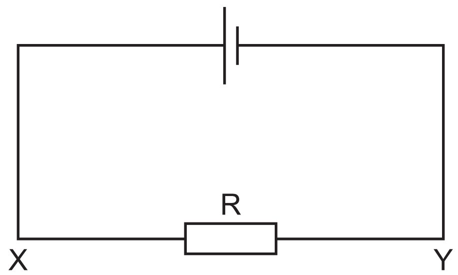
The graph shows how the electric current I through a conducting liquid varies with the potential difference V across it.

At which point on the graph does the liquid have the smallest resistance?



23 **9702/01/M/J/06/Q31**

The current in the circuit is 4.8 A.



What is the rate of flow and the direction of flow of electrons through the resistor R?

A $3.0 \times 10^{19}\,s^{-1}$ in direction X to Y

B $6.0 \times 10^{18}\,s^{-1}$ in direction X to Y

C $3.0 \times 10^{19}\,s^{-1}$ in direction Y to X

D $6.0 \times 10^{18}\,s^{-1}$ in direction Y to X

24 **9702/01/M/J/06/Q32**

Which equation is used to define resistance?

A energy = (current)^2 × resistance × time

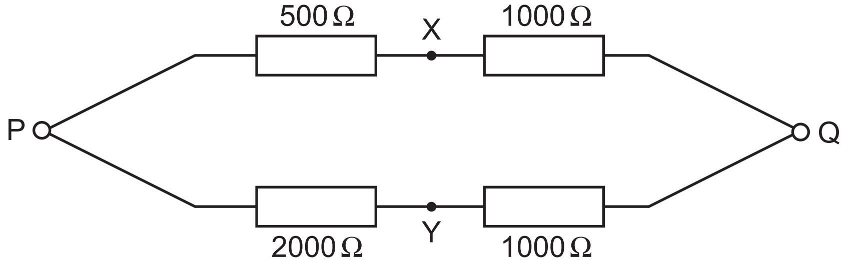
B potential difference = current × resistance

C power = (current)^2 × resistance

D resistivity = resistance × area ÷ length

25 **9702/01/M/J/06/Q33**

A p.d. of 12 V is connected between P and Q.

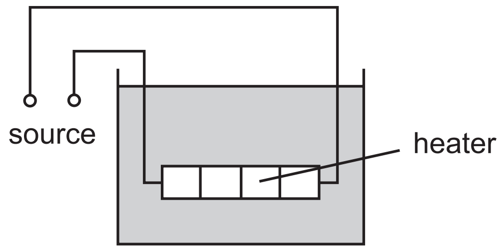


What is the p.d. between X and Y?

A 0 V **B** 4 V **C** 6 V **D** 8 V

26 **9702/01/M/J/06/Q34***

The diagram shows a low-voltage circuit for heating the water in a fish tank.



The heater has a resistance of 3.0 Ω. The voltage source has an e.m.f. of 12 V and an internal resistance of 1.0 Ω.

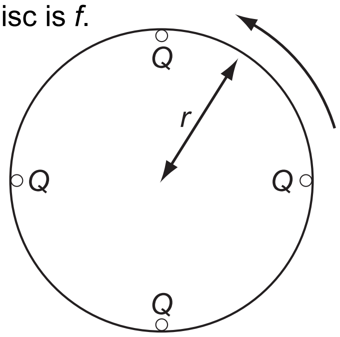
At what rate does the voltage source supply energy to the heater?

A 27 W **B** 36 W **C** 48 W **D** 64 W

27 **9702/01/O/N/06/Q31**

Four point charges, each of charge Q, are placed on the edge of an insulating disc of radius r.

The frequency of rotation of the disc is f.

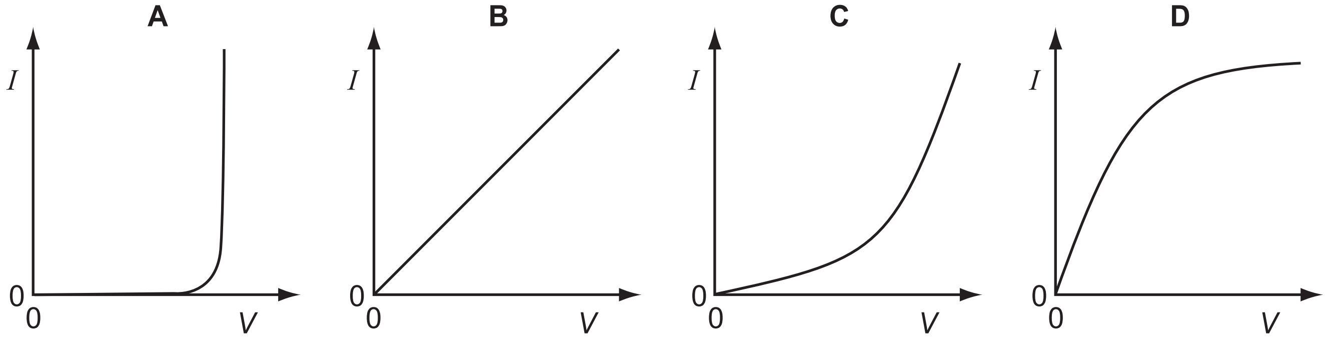


What is the equivalent electric current at the edge of the disc?

A $4Qf$ **B** $\frac{4Q}{f}$ **C** $8\pi rQf$ **D** $\frac{2Qf}{\pi r}$

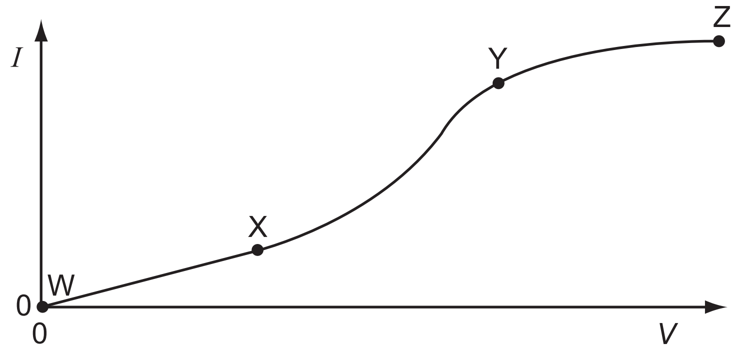
28 **9702/01/O/N/06/Q32**

Which graph shows the I – V characteristic of a filament lamp?



29 9702/01/O/N/06/Q33

An electrical component has a potential difference V across it and a current I through it. A graph of I against V is drawn and is marked in three sections WX, XY and YZ.



In which ways does the resistance of the component vary within each of the three sections?

	WX	XY	YZ
A	constant	decreases	increases
B	constant	increases	increases
C	increases	decreases	constant
D	increases	increases	decreases

30 9702/01/M/J/07/Q31

What is a correct statement of Ohm's law?

A The potential difference across a component equals the current providing the resistance and other physical conditions stay constant.

B The potential difference across a component equals the current multiplied by the resistance.

C The potential difference across a component is proportional to its resistance.

D The potential difference across a component is proportional to the current in it providing physical conditions stay constant.

31 9702/01/M/J/07/Q32

The current in a resistor is 8.0 mA.

What charge flows through the resistor in 0.020 s?

A 0.16 mC **B** 1.6 mC **C** 4.0 mC **D** 0.40 C

32 9702/01/O/N/07/Q29

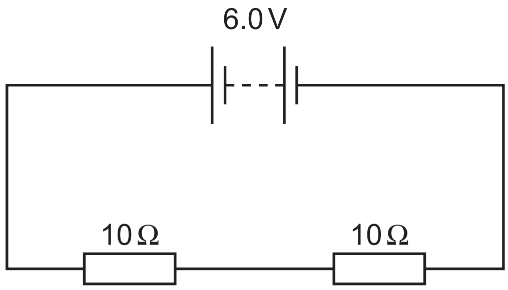
Two heating coils X and Y, of resistance R_X and R_Y respectively, deliver the same power when 12 V is applied across X and 6 V is applied across Y.

What is the ratio R_X / R_Y?

A ¼ **B** ½ **C** 2 **D** 4

33 **9702/01/O/N/07/Q30**

A battery of negligible internal resistance is connected to two 10 Ω resistors in series.



What charge flows through each of the 10 Ω resistors in 1 minute?

A 0.30 C **B** 0.60 C **C** 3.0 C **D** 18 C

34 **9702/01/O/N/07/Q31**

Two wires P and Q have resistances R_P and R_Q respectively. Wire P is twice as long as wire Q and has twice the diameter of wire Q. The wires are made of the same material.

What is the ratio $\dfrac{R_P}{R_Q}$?

A 0.5 **B** 1 **C** 2 **D** 4

35 **9702/01/M/J/08/Q32**

A power cable X has a resistance R and carries current I.

A second cable Y has a resistance $2R$ and carries current $\frac{1}{2}I$.

What is the ratio $\dfrac{\text{power dissipated in Y}}{\text{power dissipated in X}}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** 2 **D** 4

36 **9702/01/M/J/08/Q33**

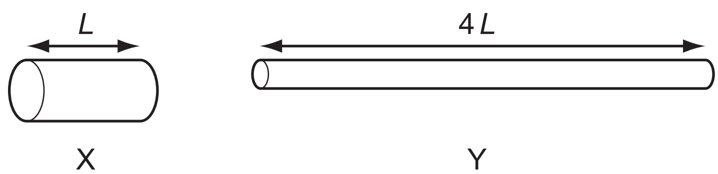
A total charge of 100 C flows through a 12 W light bulb in a time of 50 s.

What is the potential difference across the bulb during this time?

A 0.12 V **B** 2.0 V **C** 6.0 V **D** 24 V

37 **9702/01/M/J/08/Q34**

Two copper wires X and Y have the same volume. Wire Y is four times as long as wire X.



What is the ratio $\dfrac{\text{resistance of wire Y}}{\text{resistance of wire X}}$?

A 4 **B** 8 **C** 16 **D** 64

38 **9702/01/M/J/08/Q35**

The potential difference across a resistor is 12 V. The current in the resistor is 2.0 A.

4.0 C passes through the resistor.

What is the energy transferred and the time taken?

	energy / J	time / s
A	3.0	2.0
B	3.0	8.0
C	48	2.0
D	48	8.0

39 **9702/01/O/N/08/Q31***

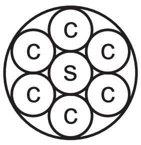
Two wires P and Q made of the same material and of the same length are connected in parallel to the same voltage supply. Wire P has diameter 2 mm and wire Q has diameter 1 mm.

What is the ratio $\frac{\text{current in P}}{\text{current in Q}}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** $\frac{2}{1}$ **D** $\frac{4}{1}$

40 **9702/01/O/N/08/Q32**

An electric power cable consists of six copper wires c surrounding a steel core s.



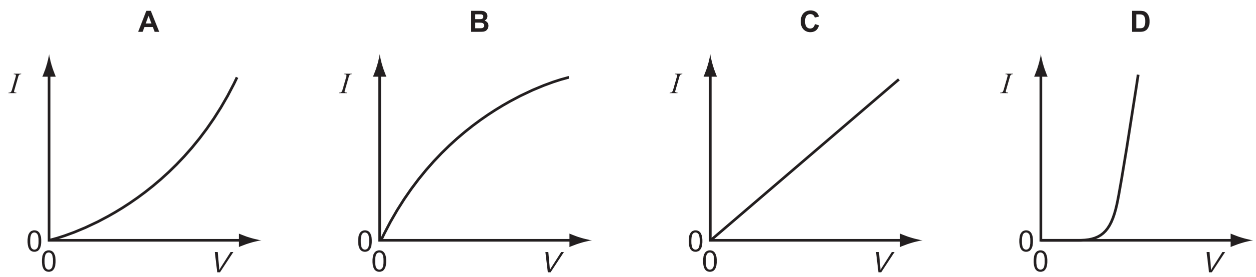
1.0 km of one of the copper wires has a resistance of 10 Ω and 1.0 km of the steel core has a resistance of 100 Ω.

What is the approximate resistance of a 1.0 km length of the power cable?

A 0.61 Ω **B** 1.6 Ω **C** 160 Ω **D** 610 Ω

41 **9702/01/O/N/08/Q33**

Which graph best represents the way the current I through a filament lamp varies with the potential difference V across it?



42 **9702/01/O/N/08/Q34**

The charge that a fully-charged 12 V car battery can supply is 100 kC. The starter motor of the car requires a current of 200 A for an average period of 2.0 s. The battery does not recharge because of a fault.

What is the maximum number of times the starter motor of the car can be used?

A 21 **B** 25 **C** 42 **D** 250

43 **9702/01/M/J/09/Q30**

Which amount of charge, flowing in the given time, will produce the largest current?

	charge / C	time / s
A	4	$\frac{1}{4}$
B	4	1
C	1	4
D	$\frac{1}{4}$	4

44 **9702/01/M/J/09/Q31**

A 12 V battery is charged for 20 minutes by connecting it to a source of electromotive force (e.m.f.). The battery is supplied with 7.2×10^4 J of energy in this time.

How much charge flows into the battery?

A 5.0 C **B** 60 C **C** 100 C **D** 6000 C

45 **9702/01/M/J/09/Q32**

What is meant by the electromotive force (e.m.f.) of a cell?

A The e.m.f. of a cell is the energy converted into electrical energy when unit charge passes through the cell.

B The e.m.f. of a cell is the energy transferred by the cell in driving unit charge through the external resistance.

C The e.m.f. of a cell is the energy transferred by the cell in driving unit charge through the internal resistance of the cell.

D The e.m.f. of a cell is the amount of energy needed to bring a unit positive charge from infinity to its positive pole.

46 **9702/12/O/N/09/Q15**

An electric railway locomotive has a maximum mechanical output power of 4.0 MW. Electrical power is delivered at 25 kV from overhead wires. The overall efficiency of the locomotive in converting electrical power to mechanical power is 80 %.

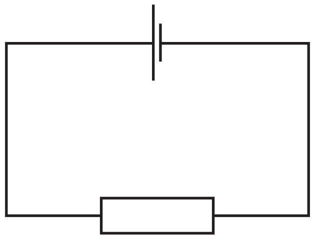
What is the current from the overhead wires when the locomotive is operating at its maximum power?

A 130 A **B** 160 A **C** 200 A **D** 250 A

47 **9702/12/O/N/09/Q29**

A cell is connected to a resistor.

At any given moment, the potential difference across the cell is less than its electromotive force.



Which statement explains this?

A The cell is continually discharging.

B The connecting wire has some resistance.

C Energy is needed to drive charge through the cell.

D Power is used when there is a current in the resistor.

48 **9702/12/O/N/09/Q30**

Which values of current and resistance will produce a rate of energy transfer of 16 J s^{-1}?

	current / A	resistance / Ω
A	1	4
B	2	8
C	4	1
D	16	1

49 **9702/12/O/N/09/Q31**

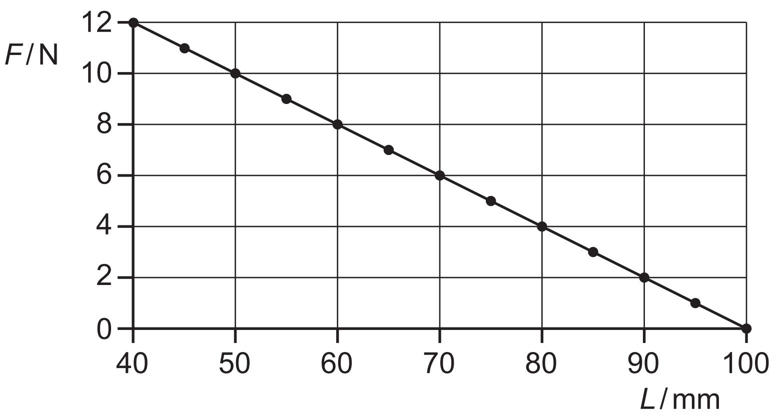
A cylindrical wire 4.0 m long has a resistance of 31 Ω and is made of metal of resistivity 1.0×10^{-6} Ω m.

What is the radius of cross-section of the wire?

A 1.0×10^{-8} m **B** 2.0×10^{-8} m **C** 6.4×10^{-8} m **D** 2.0×10^{-4} m

50 **9702/12/M/J/10/Q19**

A spring is compressed by a force. The graph shows the compressing force *F* plotted against the length *L* of the spring.

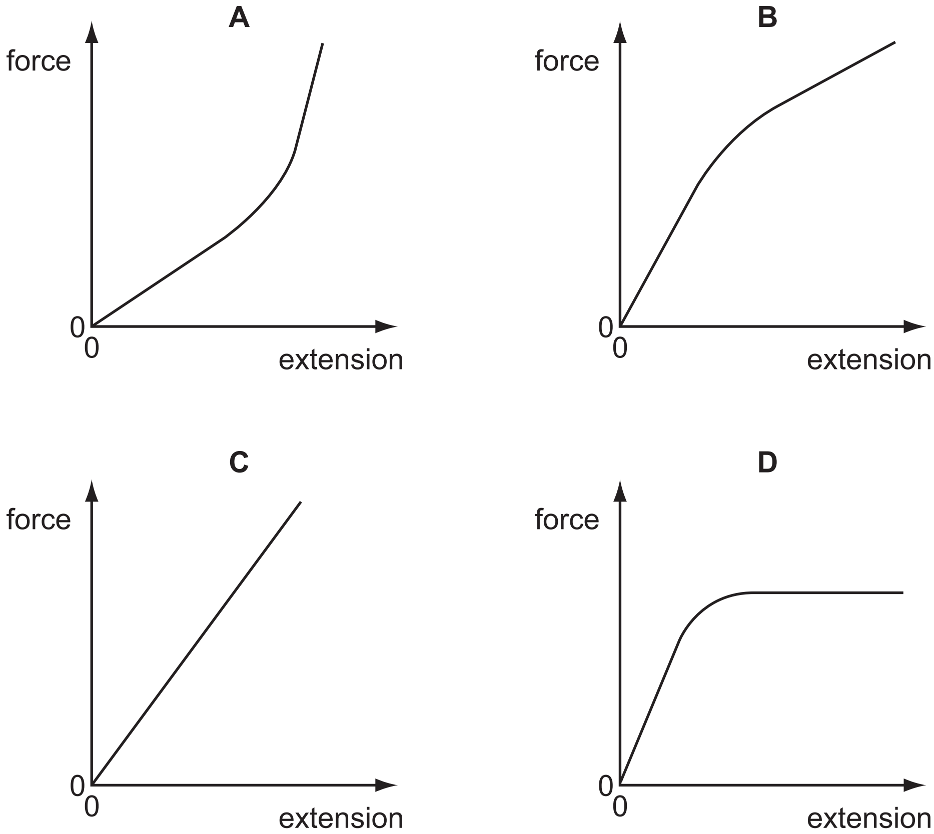


What is the spring constant of this spring?

A 0.2 N m^{-1} **B** 5 N m^{-1} **C** 100 N m^{-1} **D** 200 N m^{-1}

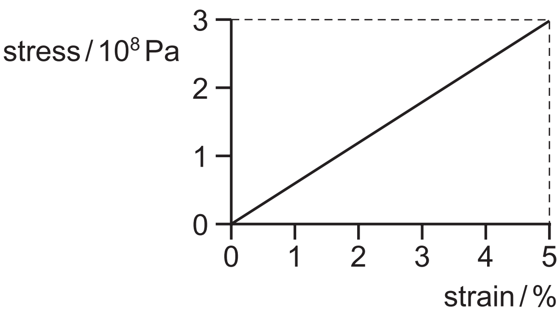
51 9702/12/M/J/10/Q20

Which graph represents the force-extension relationship of a rubber band that is stretched almost to its breaking point?



52 9702/12/M/J/10/Q21

In stress-strain experiments on metal wires, the stress axis is often marked in units of 10^8 Pa and the strain axis is marked as a percentage. This is shown for a particular wire in the diagram.



What is the value of the Young modulus for the material of the wire?

A 6.0×10^7 Pa **B** 7.5×10^8 Pa **C** 1.5×10^9 Pa **D** 6.0×10^9 Pa

53 9702/12/M/J/10/Q30

The resistance of a thermistor depends on its temperature, and the resistance of a light-dependent resistor (LDR) depends on the illumination.

Under which conditions will the resistance of both a thermistor and an LDR be highest?

	thermistor	LDR
A	highest temperature	highest illumination
B	highest temperature	lowest illumination
C	lowest temperature	highest illumination
D	lowest temperature	lowest illumination

54 9702/12/M/J/10/Q31

In terms of energy transfer W and charge q, what are the definitions of potential difference (p.d.) and electromotive force (e.m.f.)?

	p.d.	e.m.f.
A	$\frac{W}{q}$	$\frac{W}{q}$
B	$\frac{W}{q}$	Wq
C	Wq	$\frac{W}{q}$
D	Wq	Wq

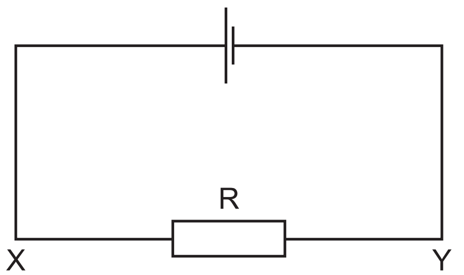
55 9702/12/M/J/10/Q36

What is the unit of resistivity?

A $\Omega\,m^{-2}$ **B** $\Omega\,m^{-1}$ **C** Ω **D** $\Omega\,m$

56 9702/11/O/N/10/Q31

The current in the circuit shown is 4.8 A.

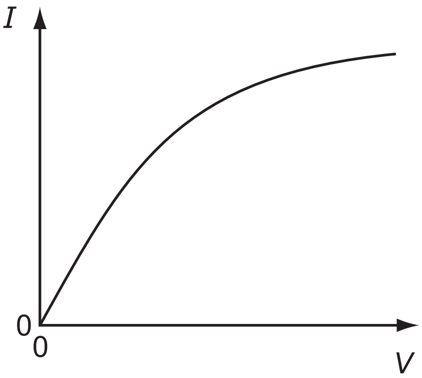


What is the direction of flow and the rate of flow of electrons through the resistor R?

	direction of flow	rate of flow
A	X to Y	$3.0 \times 10^{19}\,s^{-1}$
B	X to Y	$6.0 \times 10^{18}\,s^{-1}$
C	Y to X	$3.0 \times 10^{19}\,s^{-1}$
D	Y to X	$6.0 \times 10^{18}\,s^{-1}$

57 9702/11/O/N/10/Q32

Which component has the I-V graph shown?



A filament lamp

B light-dependent resistor

C semiconductor diode

D thermistor

58 9702/11/O/N/10/Q33

A copper wire is cylindrical and has resistance R.

What will be the resistance of a copper wire of twice the length and twice the radius?

A $\frac{R}{4}$ **B** $\frac{R}{2}$ **C** R **D** $2R$

59 9702/11/O/N/10/Q34

A relay is required to operate 800 m from its power supply. The power supply has negligible internal resistance. The relay requires 16.0 V and a current of 0.60 A to operate.

A cable connects the relay to the power supply and two of the wires in the cable are used to supply power to the relay.

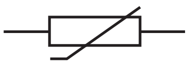
The resistance of each of these wires is 0.0050 Ω per metre.

What is the minimum output e.m.f. of the power supply?

A 16.6 V **B** 18.4 V **C** 20.8 V **D** 29.3 V

60 9702/12/O/N/10/Q30

Which electrical component is represented by the following symbol?



A a diode **B** a potentiometer **C** a resistor **D** a thermistor

61 9702/12/O/N/10/Q31

When there is **no current** in a wire, which statement about the conduction electrons in that wire is correct?

A Electrons in the wire are moving totally randomly within the wire.

B Equal numbers of electrons move at the same speed, but in opposite directions, along the wire.

C No current is flowing therefore the electrons in the wire are stationary.

D No current is flowing therefore the electrons in the wire are vibrating around a fixed point.

62 9702/12/O/N/10/Q32

A high-resistance voltmeter connected across a battery reads 6.0 V.

When the battery is connected in series with a lamp of resistance of 10 Ω, the voltmeter reading falls to 5.6 V.

Which statement explains this observation?

A The electromotive force (e.m.f.) of the battery decreases because more work is done across its internal resistance.

B The e.m.f. of the battery decreases because work is done across the lamp.

C The potential difference (p.d.) across the battery decreases because more work is done across its internal resistance.

D The p.d. across the battery decreases because work is done across the lamp.

63 9702/01/M/J/05/Q32

A copper wire of cross-sectional area 2.0 mm^2 carries a current of 10 A.

How many electrons pass through a given cross-section of the wire in one second?

A 1.0×10^{1} **B** 5.0×10^{6} **C** 6.3×10^{19} **D** 3.1×10^{25}

64 9702/11/M/J/11/Q32

A battery is marked 9.0 V.

What does this mean?

A Each coulomb of charge from the battery supplies 9.0 J of electrical energy to the whole circuit.

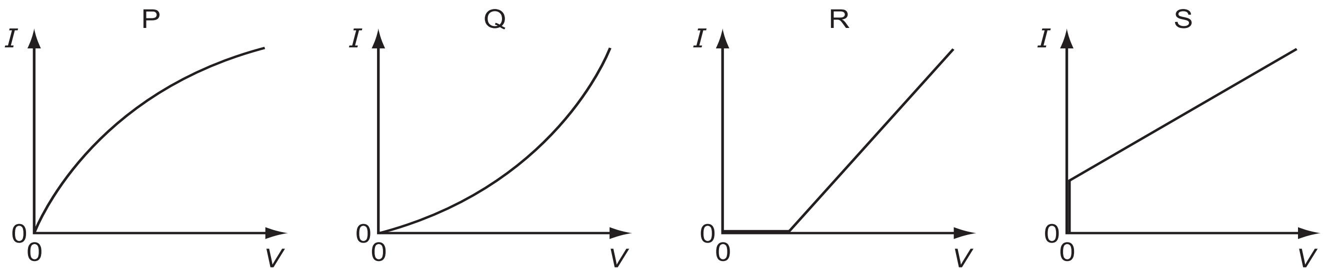
B The battery supplies 9.0 J to an external circuit for each coulomb of charge.

C The potential difference across any component connected to the battery will be 9.0 V.

D There will always be 9.0 V across the battery terminals.

65 9702/11/M/J/11/Q33

The graphs show possible current-voltage (I-V) relationships for a filament lamp and for a semiconductor diode.

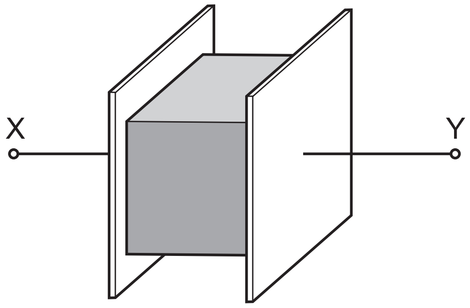


Which row best specifies the correct I-V graphs for the lamp and the diode?

	filament lamp	semiconductor diode
A	P	R
B	P	S
C	Q	R
D	Q	S

66 9702/11/M/J/11/Q34

The resistance of a metal cube is measured by placing it between two parallel plates, as shown.



The cube has volume V and is made of a material with resistivity ρ. The connections to the cube have negligible resistance.

Which expression gives the electrical resistance of the metal cube between X and Y?

A $\rho V^{\frac{1}{3}}$ **B** $\rho V^{\frac{2}{3}}$ **C** $\dfrac{\rho}{V^{\frac{1}{3}}}$ **D** $\dfrac{\rho}{V^{\frac{2}{3}}}$

67 9702/12/M/J/11/Q32

What describes the electric potential difference between two points in a wire that carries a current?

A the force required to move a unit positive charge between the points

B the ratio of the energy dissipated between the points to the current

C the ratio of the power dissipated between the points to the current

D the ratio of the power dissipated between the points to the charge moved

68 9702/12/M/J/11/Q33

A cylindrical piece of a soft, electrically-conducting material has resistance R. It is rolled out so that its length is doubled but its volume stays constant.

What is its new resistance?

A $\dfrac{R}{2}$ **B** R **C** $2R$ **D** $4R$

69 9702/12/M/J/11/Q34

A source of electromotive force (e.m.f.) E has a constant internal resistance r and is connected to an external variable resistor of resistance R.

As R is increased from a value below r to a value above r, which statement is correct?

A The terminal potential difference remains constant.

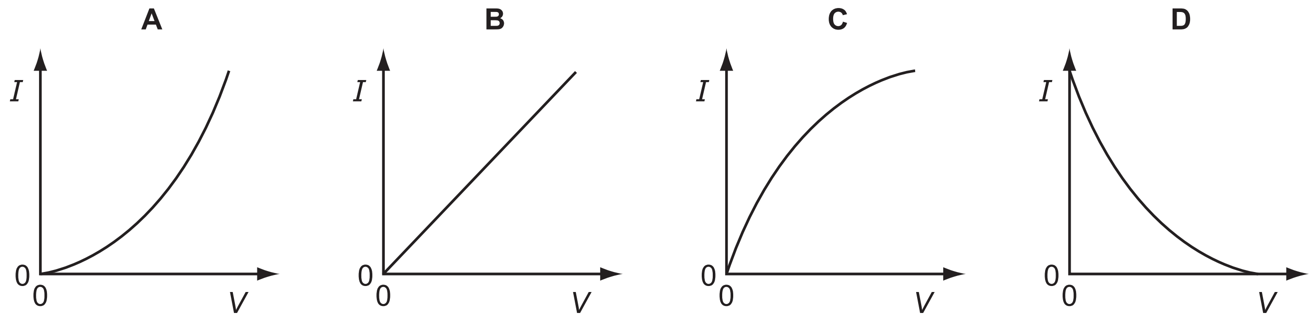
B The current in the circuit increases.

C The e.m.f. of the source increases.

D The largest output power is obtained when R reaches r.

70 9702/12/M/J/11/Q35

Which graph best represents the way in which the current I through a thermistor depends upon the potential difference V across it?



71 9702/11/O/N/11/Q31

Two horizontal parallel plate conductors are separated by a distance of 5.0 mm in air. The lower plate is earthed and the potential of the upper plate is +50 V.

What is the electric field strength E at a point midway between the plates?

A $1.0 \times 10^{4}\,\text{V m}^{-1}$ downwards

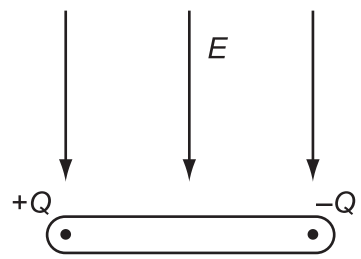
B $1.0 \times 10^{4}\,\text{V m}^{-1}$ upwards

C $2.0 \times 10^{4}\,\text{V m}^{-1}$ downwards

D $2.0 \times 10^{4}\,\text{V m}^{-1}$ upwards

72 9702/11/O/N/11/Q32

The diagram shows an insulating rod with equal and opposite point charges at each end. An electric field of strength E acts on the rod in a downwards direction.



Which row is correct?

	resultant force	resultant torque
A	zero	clockwise
B	downwards	clockwise
C	zero	anti-clockwise
D	downwards	anti-clockwise

73 9702/11/O/N/11/Q33

Which statement about electrical resistivity is correct?

A The resistivity of a material is numerically equal to the resistance in ohms of a cube of that material, the cube being of side length one metre and the resistance being measured between opposite faces.

B The resistivity of a material is numerically equal to the resistance in ohms of a one metre length of wire of that material, the area of cross-section of the wire being one square millimetre and the resistance being measured between the ends of the wire.

C The resistivity of a material is proportional to the cross-sectional area of the sample of the material used in the measurement.

D The resistivity of a material is proportional to the length of the sample of the material used in the measurement.

74 **9702/11/O/N/11/Q34**

Which of the equations that link some of the following terms is correct?

potential difference (p.d.)	V
current	I
resistance	R
charge	Q
energy	E
power	P
time	t

A $P = \dfrac{Q^2R}{t}$ **B** $ER^2 = V^2t$ **C** $\dfrac{VI}{P} = t$ **D** $PQ = EI$

75 **9702/11/O/N/11/Q35**

Which statement is **not** valid?

A Current is the speed of the charged particles that carry it.

B Electromotive force (e.m.f.) is the energy converted to electrical energy from other forms, per unit charge.

C The potential difference (p.d.) between two points is the work done in moving unit charge from one point to the other.

D The resistance between two points is the p.d. between the two points, per unit current.

76 **9702/12/O/N/11/Q32**

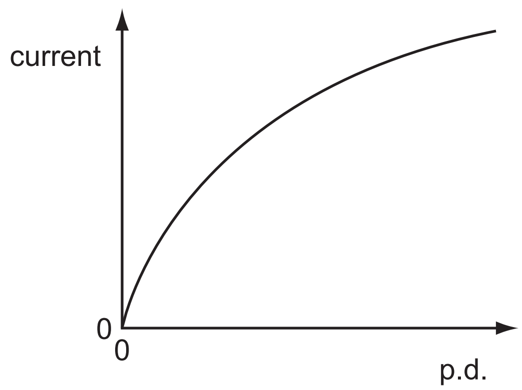
A charge of 8.0 C passes through a resistor of resistance 30 Ω at a constant rate in a time of 20 s.

What is the potential difference across the resistor?

A 0.40 V **B** 5.3 V **C** 12 V **D** 75 V

77 **9702/12/O/N/11/Q33**

The graph shows the variation with potential difference (p.d.) of the current in a lamp filament.



Which statement explains the shape of this graph?

A As the filament temperature rises, electrons can pass more easily through the filament.

B It takes time for the filament to reach its working temperature.

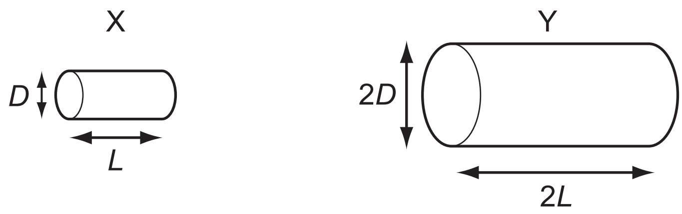
C The power output of the filament is proportional to the square of the current in it.

D The resistance of the filament increases with a rise in temperature.

78 9702/12/O/N/11/Q34

Two electrically-conducting cylinders X and Y are made from the same material.

Their dimensions are as shown.



The resistance of each cylinder is measured between its ends.

What is the ratio $\dfrac{\text{resistance of X}}{\text{resistance of Y}}$?

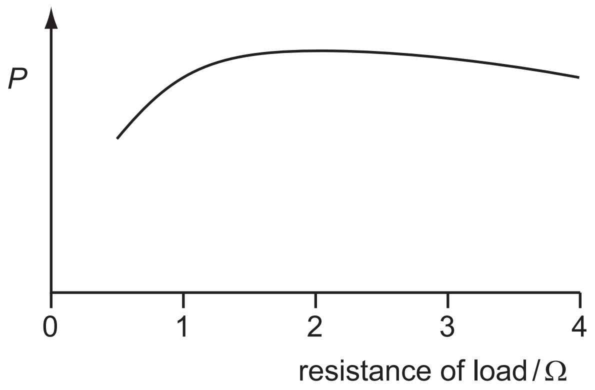
A $\dfrac{2}{1}$ **B** $\dfrac{1}{1}$ **C** $\dfrac{1}{2}$ **D** $\dfrac{1}{4}$

79 9702/12/O/N/11/Q35

A power supply of electromotive force (e.m.f.) 12 V and internal resistance 2 Ω is connected in series with a load resistor. The value of the load resistor is varied from 0.5 Ω to 4 Ω.

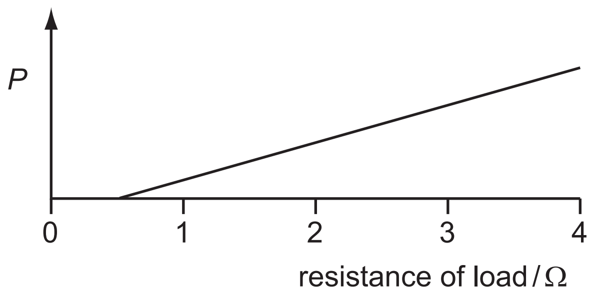
Which graph shows how the power P dissipated in the load resistor varies with the resistance of the load resistor?

A



B

C



D

80 **9702/12/M/J/12/Q32**

When will 1 C of charge pass a point in an electrical circuit?

A when 1 A moves through a potential difference of 1 V

B when a power of 1 W is used for 1 s

C when the current is 5 mA for 200 s

D when the current is 10 A for 10 s

81 **9702/12/M/J/12/Q33**

Two copper wires of the same length but different diameters carry the same current.

Which statement about the flow of charged particles through the wires is correct?

A Charged particles are provided by the power supply. Therefore the speed at which they travel depends only on the voltage of the supply.

B The charged particles in both wires move with the same average speed because the current in both wires is the same.

C The charged particles move faster through the wire with the larger diameter because there is a greater volume through which to flow.

D The charged particles move faster through the wire with the smaller diameter because it has a larger potential difference applied to it.

82 **9702/12/M/J/12/Q34**

A power cable X has resistance R and carries current I.

A second cable Y has resistance $2R$ and carries current $\frac{1}{2}I$.

What is the ratio $\dfrac{\text{power dissipated in Y}}{\text{power dissipated in X}}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** 2 **D** 4

83 **9702/13/M/J/12/Q32**

An iron wire has length 8.0 m and diameter 0.50 mm. The wire has resistance R.

A second iron wire has length 2.0 m and diameter 1.0 mm.

What is the resistance of the second wire?

A $\dfrac{R}{16}$ **B** $\dfrac{R}{8}$ **C** $\dfrac{R}{2}$ **D** R

84 **9702/12/M/J/13/Q32**

A power cable has length 2000 m. The cable is made of twelve parallel strands of copper wire, each with diameter 0.51 mm.

What is the resistance of the cable? (resistivity of copper = $1.7 \times 10^{-8}\,\Omega\,m$)

A 0.014 Ω **B** 3.5 Ω **C** 14 Ω **D** 166 Ω

85 **9702/12/M/J/13/Q34**

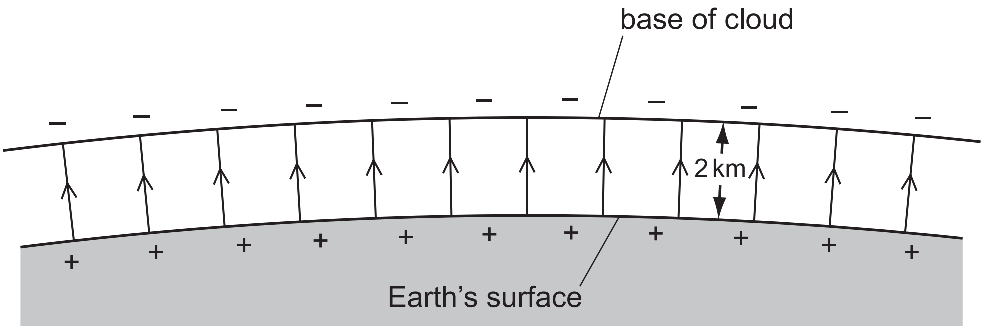
A filament lamp has a resistance of 180 Ω when the current in it is 500 mA.

What is the power transformed in the lamp?

A 45 W **B** 50 W **C** 90 W **D** 1400 W

86 **9702/13/M/J/12/Q33**

Lightning can occur between a charged cloud and the Earth's surface when the electric field strength in the intervening atmosphere reaches $25\,\text{kN}\,\text{C}^{-1}$. The diagram shows the electric field between the base of a cloud and the Earth's surface.



What is the minimum potential difference between the Earth and the base of a cloud, 2 km high, for lightning to occur?

A 12.5 MV **B** 25 MV **C** 50 MV **D** 100 MV

87 **9702/13/M/J/12/Q34**

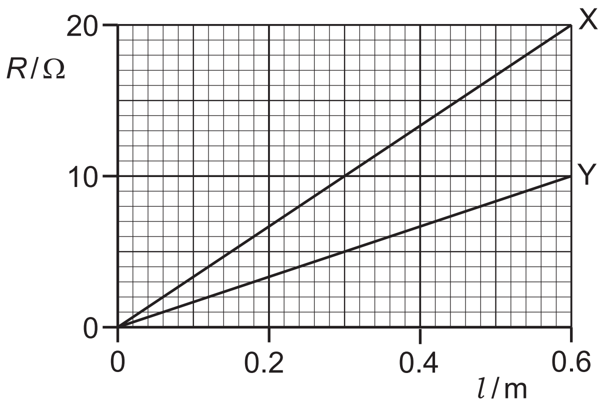
There is a current of 10 mA in a conductor for half an hour.

How much charge passes a point in the conductor in this time?

A 0.3 C **B** 5 C **C** 18 C **D** 300 C

88 **9702/12/O/N/12/Q34**

The graph shows the variation with length l of resistance R for two wires X and Y made from the same material.



What does the graph show?

A cross-sectional area of X = 2 × cross-sectional area of Y

B resistivity of X = 2 × resistivity of Y

C when equal lengths of X and Y are connected in series to a battery,
power in X = 2 × power in Y

D when equal lengths of X and Y are connected in parallel to a battery,
current in X = 2 × current in Y

89 9702/12/O/N/12/Q35

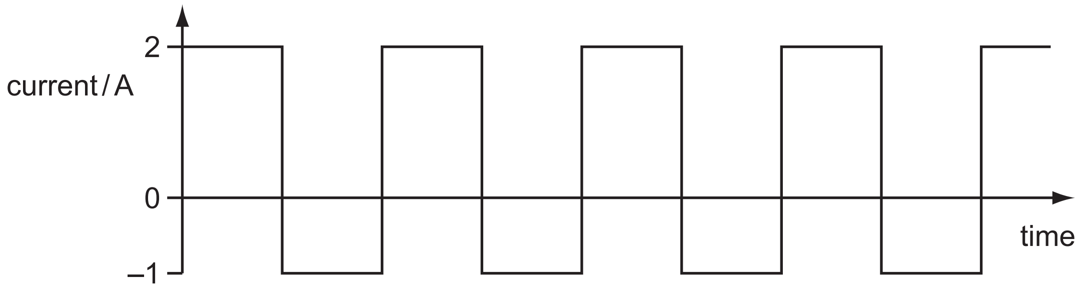
A cell of internal resistance 2.0 Ω and electromotive force (e.m.f.) 1.5 V is connected to a resistor of resistance 3.0 Ω.

What is the potential difference across the 3.0 Ω resistor?

A 1.5 V **B** 1.2 V **C** 0.9 V **D** 0.6 V

90 9702/12/O/N/12/Q36

A 100 Ω resistor conducts a current with changing direction and magnitude, as shown.



What is the mean power dissipated in the resistor?

A 100 W **B** 150 W **C** 250 W **D** 400 W

91 9702/11/M/J/17/Q33

Which values of current and resistance will produce a rate of energy transfer of $16\,J\,s^{-1}$?

	current / A	resistance / Ω
A	1	4
B	2	8
C	4	1
D	16	1

92 9702/13/O/N/12/Q33

A copper wire is stretched so that its diameter is reduced from 1.0 mm to a uniform 0.5 mm.

The resistance of the unstretched copper wire is 0.2 Ω.

What will be the resistance of the stretched wire?

A 0.4 Ω **B** 0.8 Ω **C** 1.6 Ω **D** 3.2 Ω

93 9702/12/M/J/13/Q33

A low-voltage supply with an e.m.f. of 20 V and an internal resistance of 1.5 Ω is used to supply power to a heater of resistance 6.5 Ω in a fish tank.

What is the power supplied to the water in the fish tank?

A 41 W **B** 50 W **C** 53 W **D** 62 W

94 9702/13/O/N/12/Q34

Four statements about potential difference or electromotive force are listed.

1 It involves changing electrical energy into other forms.

2 It involves changing other energy forms into electrical energy.

3 It is the energy per unit charge to move charge right round a circuit.

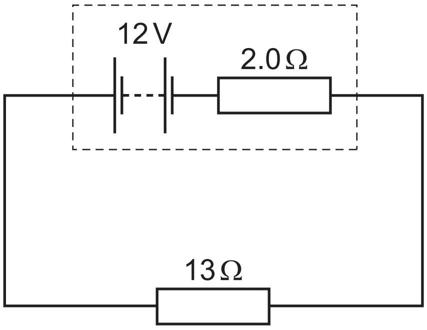
4 It is the work done per unit charge by the charge moving from one point to another.

Which statements apply to potential difference and which apply to electromotive force?

	potential difference	electromotive force
A	1 and 3	2 and 4
B	1 and 4	2 and 3
C	2 and 3	1 and 4
D	2 and 4	1 and 3

95 9702/11/M/J/13/Q32

A power supply of electromotive force (e.m.f.) 12 V and internal resistance 2.0 Ω is connected in series with a 13 Ω resistor.



What is the power dissipated in the 13 Ω resistor?

A 8.3 W **B** 9.6 W **C** 10 W **D** 11 W

96 9702/11/M/J/13/Q33

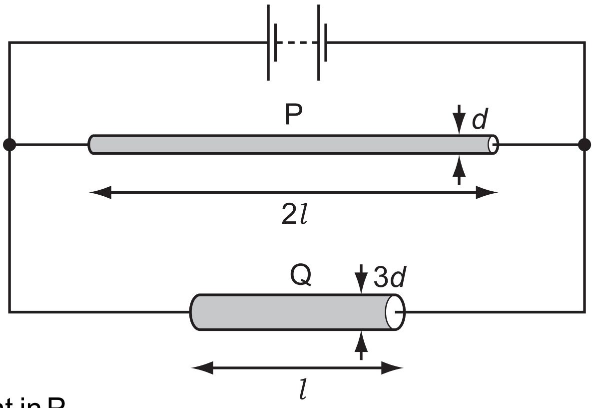
When a battery is connected to a resistor, the battery gradually becomes warm. This causes the internal resistance of the battery to increase whilst its e.m.f. stays unchanged.

As the internal resistance of the battery increases, how do the terminal potential difference and the output power change, if at all?

	terminal potential difference	output power
A	decrease	decrease
B	decrease	unchanged
C	unchanged	decrease
D	unchanged	unchanged

97 **9702/12/M/J/13/Q35**

Two wires P and Q made of the same material are connected to the same electrical supply. P has twice the length of Q and one-third of the diameter of Q, as shown in the diagram.



What is the ratio $\dfrac{\text{current in P}}{\text{current in Q}}$?

A $\dfrac{2}{3}$ **B** $\dfrac{2}{9}$ **C** $\dfrac{1}{6}$ **D** $\dfrac{1}{18}$

98 **9702/13/M/J/13/Q21**

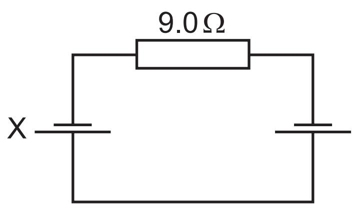
A spring is stretched over a range within which elastic deformation occurs. Its spring constant is $3.0\,\text{N}\,\text{cm}^{-1}$.

Which row, for the stated applied force, gives the correct extension and strain energy?

	force / N	extension / cm	strain energy / mJ
A	3.0	1.0	1.5
B	6.0	2.0	120
C	12.0	3.0	180
D	24.0	8.0	960

99 **9702/13/M/J/13/Q31**

Two cells X and Y are connected in series with a resistor of resistance 9.0 Ω, as shown.



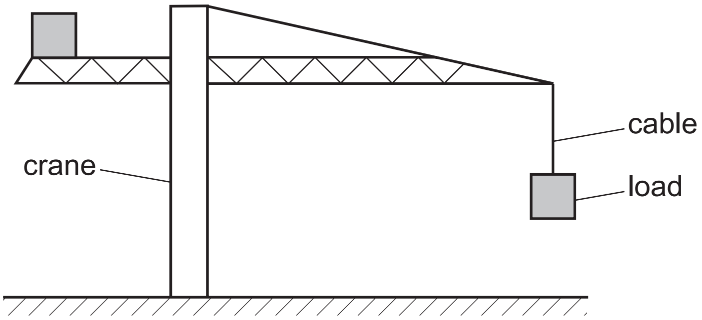
Cell X has an electromotive force (e.m.f.) of 1.0 V and an internal resistance of 1.0 Ω. Cell Y has an e.m.f. of 2.0 V and an internal resistance of 2.0 Ω.

What is the current in the circuit?

A 0.25 A **B** 0.17 A **C** 0.10 A **D** 0.083 A

100 9702/12/M/J/13/Q23

The diagram shows a large crane on a construction site lifting a cube-shaped load.



A model is made of the crane, its load and the cable supporting the load.

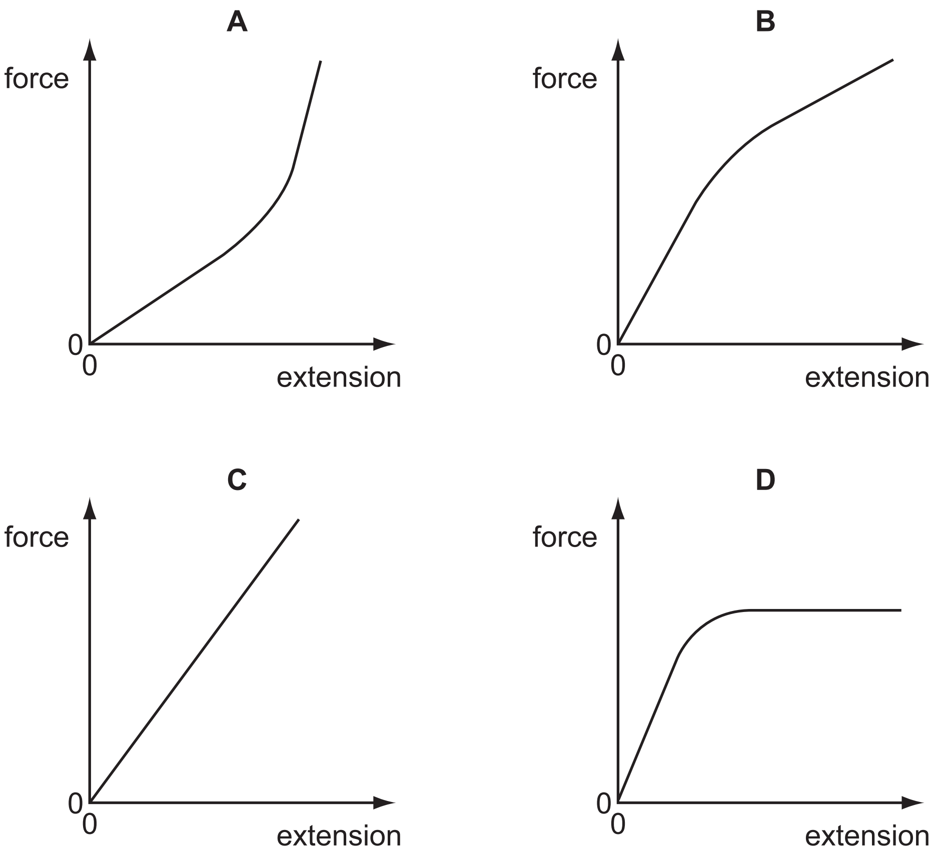
The material used for each part of the model is the same as that in the full-size crane, cable and load. The model is one tenth full-size in all linear dimensions.

What is the ratio $\dfrac{\text{extension of the cable on the full-size crane}}{\text{extension of the cable on the model crane}}$?

A 10^0 **B** 10^1 **C** 10^2 **D** 10^3

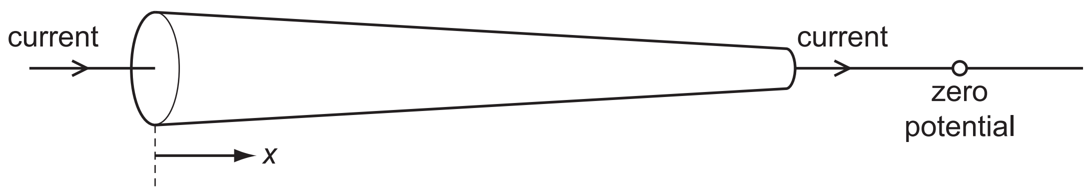
101 9702/13/M/J/13/Q20

Which graph represents the force-extension relationship of a rubber band that is stretched almost to its breaking point?

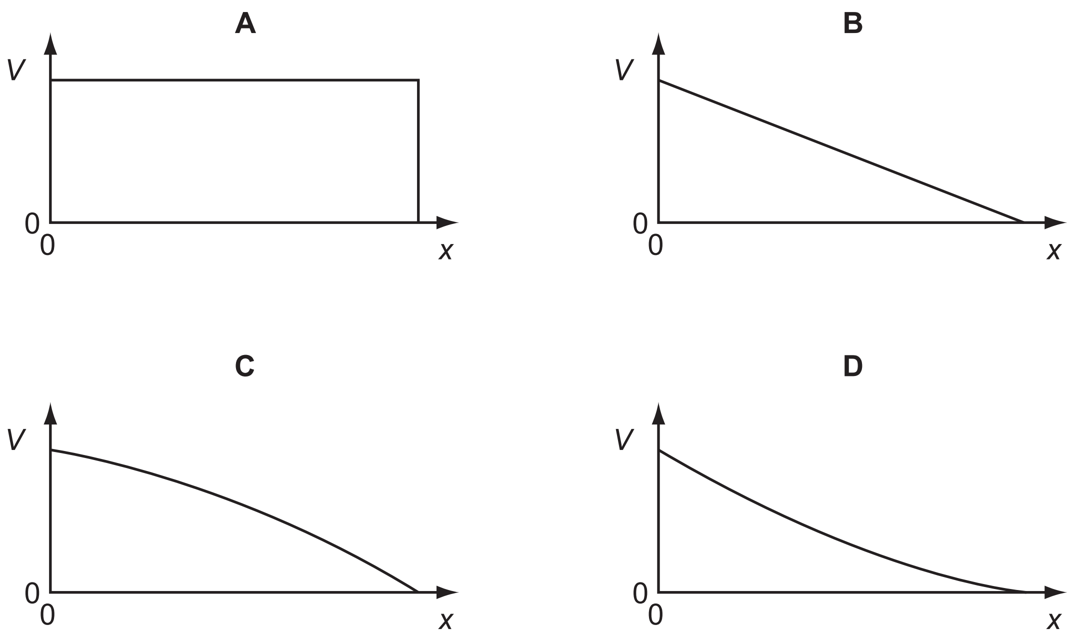


102 **9702/13/M/J/13/Q32**

The circular cross-sectional area of a metal wire varies along its length. There is a current in the wire. The narrow end of the wire is at a reference potential of zero.

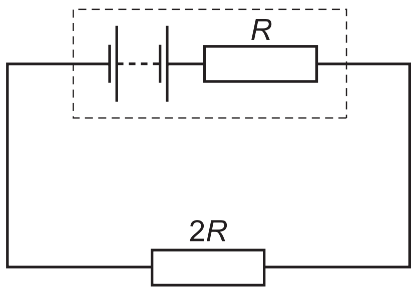


Which graph best represents the variation with distance x along the wire of the potential difference V relative to the reference zero?



103 **9702/12/O/N/13/Q33**

The diagram shows an electric circuit in which the resistance of the external resistor is 2R and the internal resistance of the source is R.



What is the ratio $\dfrac{\text{power in external resistor}}{\text{power in internal resistance}}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** 2 **D** 4

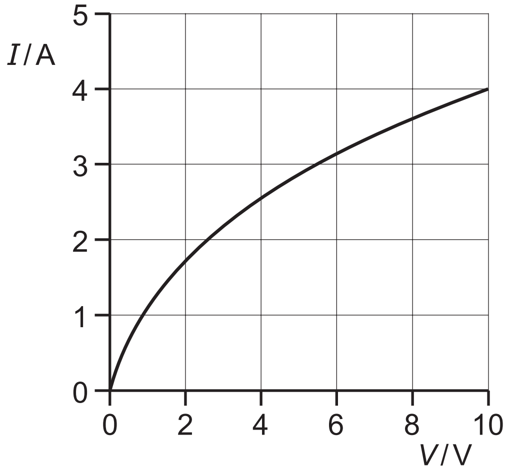
104 9702/13/M/J/13/Q33

The graph shows how current I varies with voltage V for a filament lamp.

Since the graph is not a straight line, the resistance of the lamp varies with V.

Which row gives the correct resistance at the stated value of V?

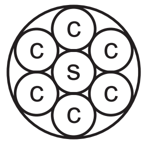
	V/V	R/Ω
A	2.0	1.5
B	4.0	3.2
C	6.0	1.9
D	8.0	0.9



105 9702/13/M/J/13/Q34

An electric power cable consists of six copper wires c surrounding a steel core s.

A length of 1.0 km of one of the copper wires has a resistance of 10 Ω and 1.0 km of the steel core has a resistance of 100 Ω.

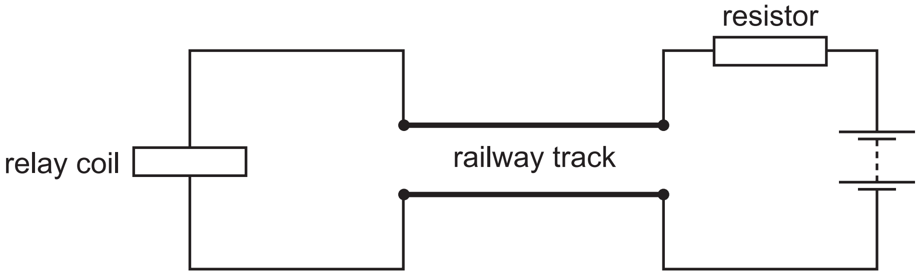


What is the approximate resistance of a 1.0 km length of the power cable?

A 0.61 Ω **B** 1.6 Ω **C** 160 Ω **D** 610 Ω

106 9702/13/M/J/13/Q35

The diagram shows a length of track from a model railway connected to a battery, a resistor and a relay coil.



With no train present, there is a current in the relay coil which operates a switch to turn on a light.

When a train occupies the section of track, most of the current flows through the wheels and axles of the train in preference to the relay coil. The switch in the relay turns off the light.

Why is a resistor placed between the battery and the track?

A to limit the heating of the wheels of the train

B to limit the energy lost in the relay coil when a train is present

C to prevent a short circuit of the battery when a train is present

D to protect the relay when a train is present

107 9702/12/O/N/13/Q34

Two lamps are connected in series to a 250 V power supply. One lamp is rated 240 V, 60 W and the other is rated 10 V, 2.5 W.

Which statement most accurately describes what happens?

A Both lamps light at less than their normal brightness.

B Both lamps light normally.

C Only the 60 W lamp lights.

D The 10 V lamp blows.

108 9702/12/O/N/13/Q35

The wire of a heating element has resistance R. The wire breaks and is replaced by a different wire.
Data for the original wire and the replacement wire are shown in the table.

	length	diameter	resistivity of metal
original wire	l	d	ρ
replacement wire	l	$2d$	2ρ

What is the resistance of the replacement wire?

A $\frac{R}{4}$ **B** $\frac{R}{2}$ **C** R **D** $2R$

109 9702/13/O/N/13/Q32

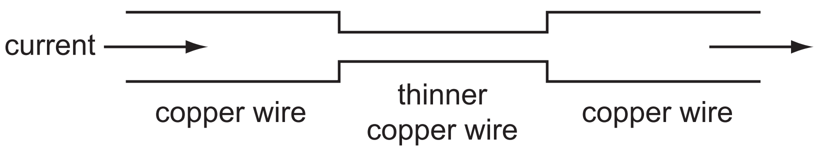
The current in a component is reduced uniformly from 100 mA to 20 mA over a period of 8.0 s.

What is the charge that flows during this time?

A 160 mC **B** 320 mC **C** 480 mC **D** 640 mC

110 9702/13/O/N/13/Q33

An electric current is passed from a thick copper wire through a section of thinner copper wire before entering a second thick copper wire as shown.



Which statement about the current and the speed of electrons in the wires is correct?

A The current and the speed of the electrons in the thinner wire are both less than in the thicker copper wires.

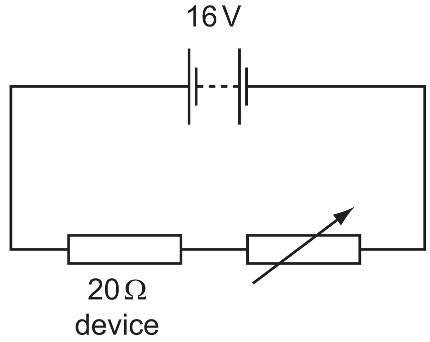
B The current and the speed of the electrons is the same in all the wires.

C The current is the same in all the wires but the speed of the electrons in the thinner wire is greater than in the thicker wires.

D The current is the same in all the wires but the speed of the electrons in the thinner wire is less than in the thicker wire.

111 9702/13/O/N/13/Q34

An electrical device of fixed resistance 20 Ω is connected in series with a variable resistor and a battery of electromotive force (e.m.f.) 16 V and negligible internal resistance.



What is the resistance of the variable resistor when the power dissipated in the electrical device is 4.0 W?

A 16 Ω **B** 36 Ω **C** 44 Ω **D** 60 Ω

112 9702/13/O/N/13/Q35

A copper wire is cylindrical and has resistance R.

What will be the resistance of a copper wire of twice the length and twice the radius?

A $\frac{R}{4}$ **B** $\frac{R}{2}$ **C** R **D** $2R$

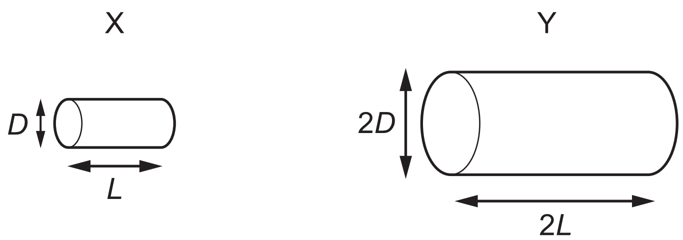
113 9702/11/M/J/14/Q30

Two electrically-conducting cylinders X and Y are made from the same material.

Their dimensions are as shown.

The resistance between the ends of each cylinder is measured.

What is the ratio $\frac{\text{resistance of X}}{\text{resistance of Y}}$?

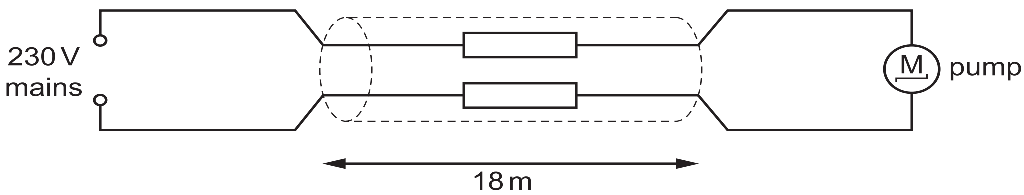


A $\frac{2}{1}$ **B** $\frac{1}{1}$

C $\frac{1}{2}$ **D** $\frac{1}{4}$

114 9702/12/M/J/14/Q32

The diagram shows an electric pump for a garden fountain connected by an 18 m cable to a 230 V mains electrical supply.



The performance of the pump is acceptable if the potential difference (p.d.) across it is at least 218 V. The current through it is then 0.83 A.

What is the maximum resistance per metre of each of the two wires in the cable if the pump is to perform acceptably?

A 0.40 Ω m^{-1} **B** 0.80 Ω m^{-1} **C** 1.3 Ω m^{-1} **D** 1.4 Ω m^{-1}

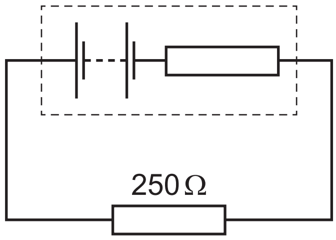
115 9702/11/M/J/14/Q31

A battery, with a constant internal resistance, is connected to a resistor of resistance 250 Ω, as shown.

The current in the resistor is 40 mA for a time of 60 s. During this time 6.0 J of energy is lost in the internal resistance.

What are the energy supplied to the external resistor during the 60 s and the e.m.f. of the battery?

	energy / J	e.m.f. / V
A	2.4	2.4
B	2.4	7.5
C	24	10.0
D	24	12.5



116 9702/11/M/J/14/Q35

The potential difference across a component in a circuit is 2.0 V.

How many electrons must flow through this component in order for it to be supplied with 4.8 J of energy?

A 2.6×10^{18} **B** 1.5×10^{19} **C** 3.0×10^{19} **D** 6.0×10^{19}

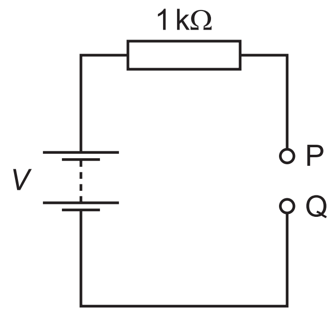
117 9702/12/M/J/14/Q30

A battery of electromotive force (e.m.f.) *V* and negligible internal resistance is connected to a 1 kΩ resistor, as shown.

A student attempts to measure the potential difference (p.d.) between points P and Q using two voltmeters, one at a time. The first voltmeter has a resistance of 1 kΩ and the second voltmeter has a resistance of 1 MΩ.

What are the readings of the voltmeters?

	reading on voltmeter with 1 kΩ resistance	reading on voltmeter with 1 MΩ resistance
A	$\frac{V}{2}$	$\frac{V}{2}$
B	$\frac{V}{2}$	V
C	V	$\frac{V}{2}$
D	V	V



118 9702/12/M/J/14/Q31

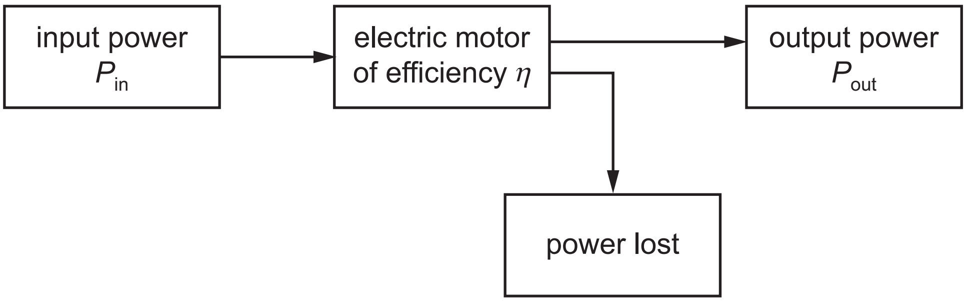
A copper wire is to be replaced by an aluminium alloy wire of the same length and resistance. Copper has half the resistivity of the alloy.

What is the ratio $\dfrac{\text{diameter of alloy wire}}{\text{diameter of copper wire}}$?

A $\sqrt{2}$ **B** 2 **C** $2\sqrt{2}$ **D** 4

119 9702/13/M/J/14/Q16

An electric motor has an input power P_{in}, useful output power P_{out} and efficiency η.



How much power is lost by the motor?

A ηP_{in} **B** $\left(\frac{1}{\eta}-1\right)P_{in}$ **C** ηP_{out} **D** $\left(\frac{1}{\eta}-1\right)P_{out}$

120 9702/13/M/J/14/Q33

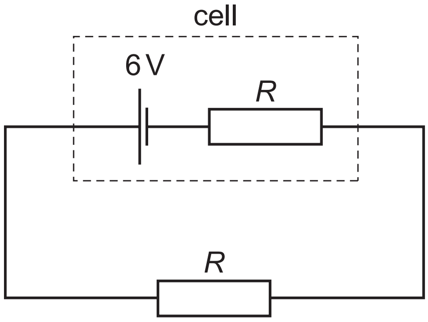
Two wires P and Q made of the same material and of the same length are connected in parallel to the same voltage supply. Wire P has diameter 2 mm and wire Q has diameter 1 mm.

What is the ratio $\frac{\text{current in P}}{\text{current in Q}}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** $\frac{2}{1}$ **D** $\frac{4}{1}$

121 9702/13/M/J/14/Q34

A cell has an electromotive force (e.m.f.) of 6 V and internal resistance R. An external resistor, also of resistance R, is connected across this cell, as shown.



Power P is dissipated by the external resistor.

The cell is replaced by a different cell that has an e.m.f. of 6 V and negligible internal resistance.

What is the new power that is dissipated in the external resistor?

A $0.5P$ **B** P **C** $2P$ **D** $4P$

122 9702/11/O/N/14/Q32

A pencil is used to draw a line of length 30 cm and width 1.2 mm. The resistivity of the material in the pencil is $2.0 \times 10^{-5}\,\Omega\,\text{m}$ and the resistance of the line is 40 kΩ.

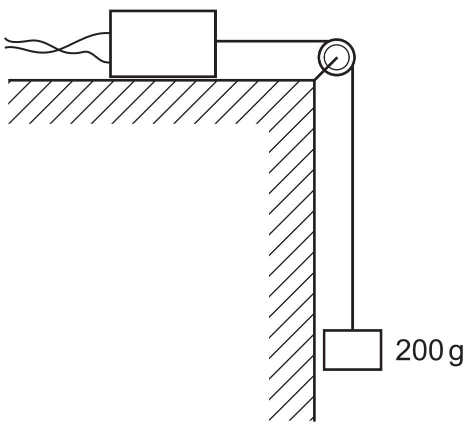
What is the thickness of the line?

A 1.25×10^{-10} m **C** 1.25×10^{-7} m

B 1.25×10^{-8} m **D** 1.25×10^{-5} m

123 9702/11/O/N/14/Q15

A small electric motor is mounted on a bench, as shown. The motor is connected to a 6.0 V supply and the current in the motor is 0.50 A. The motor is 50% efficient.

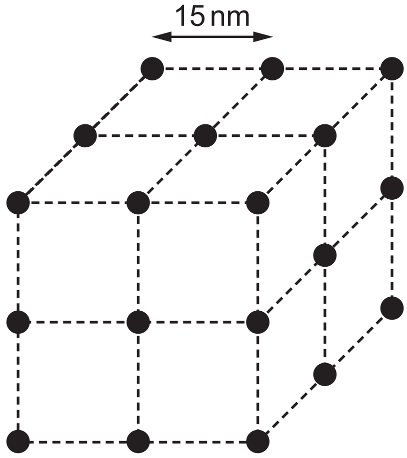


What is the time taken to lift a mass of 200 g up through a height of 90 cm?

A 0.59 s **B** 0.85 s **C** 1.2 s **D** 2.7 s

124 9702/11/O/N/14/Q19

The diagram shows the atoms of a substance with the atoms at the corners of a cube. The average separation of the atoms at a particular temperature is 15 nm.



When the temperature changes so that the average separation becomes 17 nm, by which factor will the density of the substance change?

A 0.61 **B** 0.69 **C** 0.78 **D** 0.88

125 9702/13/M/J/15/Q35

A source of e.m.f. 9.0 mV has an internal resistance of 6.0 Ω.

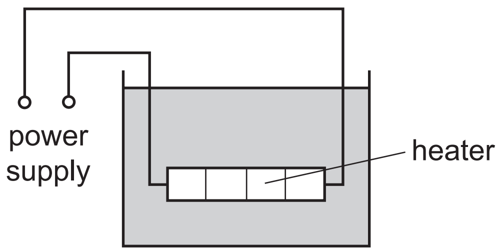
It is connected across a galvanometer of resistance 30 Ω.

What is the current in the galvanometer?

A 250 μA **B** 300 μA **C** 1.5 mA **D** 2.5 mA

126 **9702/01/M/J/06/Q34**

The diagram shows a low-voltage circuit for heating the water in a fish tank.



The heater has a resistance of 3.0 Ω. The power supply has an e.m.f. of 12 V and an internal resistance of 1.0 Ω.

At which rate is energy supplied to the heater?

A 27 W **B** 36 W **C** 48 W **D** 64 W

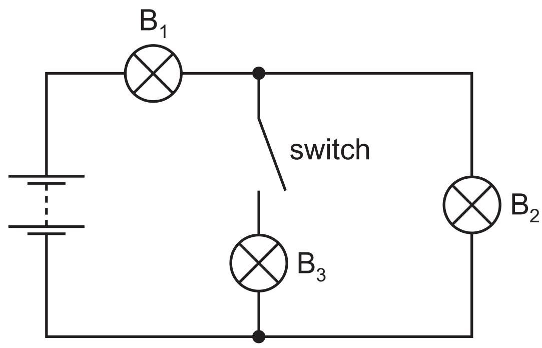
127 **9702/11/O/N/14/Q30**

B_1, B_2 and B_3 are three identical lamps. They are connected to a battery with zero internal resistance, as shown.

Initially the switch is closed. The switch is then opened and lamp B_3 goes out.

What happens to the brightness of lamps B_1 and B_2 when the switch is opened?

	brightness of lamp B_1	brightness of lamp B_2
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases



128 **9702/11/O/N/14/Q31**

A battery is marked 9.0 V.

What does this mean?

A Each coulomb of charge from the battery supplies 9.0 J of electrical energy to the whole circuit.

B The battery supplies 9.0 J to an external circuit for each coulomb of charge.

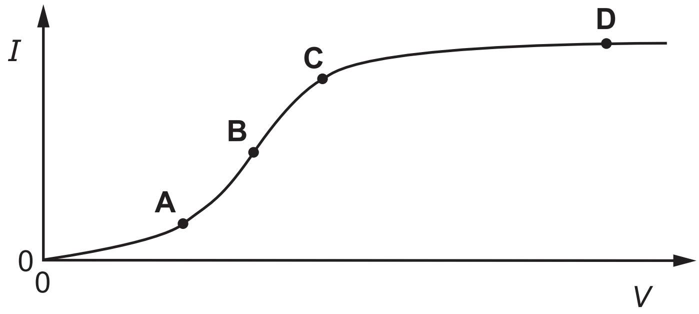
C The potential difference across any component connected to the battery will be 9.0 V.

D There will always be 9.0 V across the battery terminals.

129 9702/11/O/N/14/Q34

The graph shows how the electric current I through a conducting liquid varies with the potential difference V across it.

At which point on the graph does the liquid have the smallest resistance?



130 9702/12/O/N/14/Q35

The combined resistance R_T of two resistors of resistances R_1 and R_2 connected in parallel is given by the formula shown.

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

Which statement is used in the derivation of this formula?

A The currents through the two resistors are equal.

B The potential difference across each resistor is the same.

C The supply current is split between the two resistors in the same ratio as the ratio of their resistances.

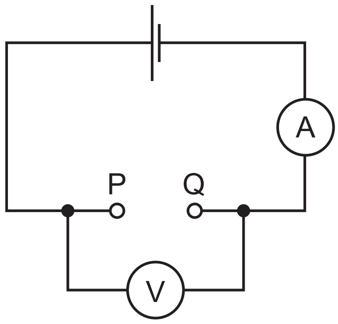
D The total power dissipated is the sum of the powers dissipated in the two resistors separately.

131 9702/13/O/N/14/Q34

A student found two unmarked resistors. To determine the resistance of the resistors, the circuit below was set up. The resistors were connected in turn between P and Q, noting the current readings. The voltage readings were noted without the resistors and with each resistor in turn.

The results were entered into a spreadsheet as shown.

1.5	1.3	28	46
1.5	1.4	14	100



The student forgot to enter the column headings.

Which order of the headings would be correct?

A

e.m.f. / V	V / V	R / Ω	I / mA

B

V / V	e.m.f. / V	R / Ω	I / mA

C

V / V	e.m.f. / V	I / mA	R / Ω

D

e.m.f. / V	V / V	I / mA	R / Ω

132 9702/13/O/N/14/Q33

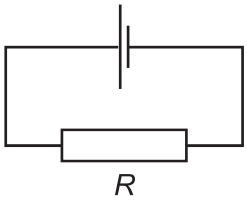
A metal wire of length 0.50 m has a resistance of 12 Ω.

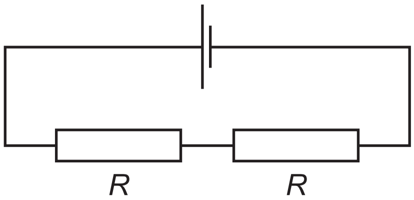
What is the resistance of a wire of length 2.0 m and made of the same material, but with half the diameter?

A 12 Ω **B** 48 Ω **C** 96 Ω **D** 192 Ω

133 9702/11/M/J/15/Q33

The diagrams show two different circuits.





The cells in each circuit have the same electromotive force and zero internal resistance. The three resistors each have the same resistance R.

In the circuit on the left, the power dissipated in the resistor is P.

What is the total power dissipated in the circuit on the right?

A $\frac{P}{4}$ **B** $\frac{P}{2}$ **C** P **D** $2P$

134 9702/11/M/J/15/Q34

Which equation that links some of the following terms is correct?

potential difference (p.d.)	V
current	I
resistance	R
charge	Q
energy	E
power	P
time	t

A $P = \frac{Q^2R}{t}$

B $ER^2 = V^2t$

C $\frac{VI}{P} = t$

D $PQ = EI$

135 9702/11/M/J/15/Q35

The charge that an electric battery can deliver is specified in ampere-hours.

For example, a battery of capacity 40 ampere-hours could supply, when fully charged, 0.2 A for 200 hours.

What is the maximum energy that a fully charged 12 V, 40 ampere-hour battery could supply?

A 1.7 kJ **B** 29 kJ **C** 1.7 MJ **D** 29 MJ

136 9702/12/M/J/15/Q31

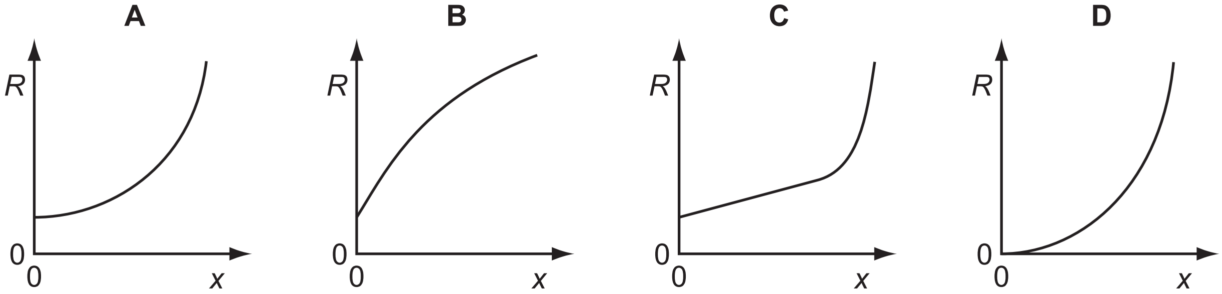
Which unit is **not** used in either the definition of the coulomb or the definition of the volt?

A ampere **B** joule **C** ohm **D** second

137 **9702/12/M/J/15/Q32**

When a thin metal wire is stretched, it becomes longer and thinner. This causes a change in the resistance of the wire. The volume of the wire remains constant.

Which graph could represent the variation with extension *x* of the resistance *R* of the wire?



138 **9702/12/M/J/15/Q33**

Which statement is **not** valid?

A Current is the speed of the charged particles that carry it.

B Electromotive force (e.m.f.) is the energy converted to electrical energy from other forms per unit charge.

C The potential difference (p.d.) between two points is the work done per unit charge when moving charge from one point to the other.

D The resistance between two points is the p.d. between the two points per unit current.

139 **9702/12/M/J/15/Q34**

A cell of e.m.f. *E* delivers a charge *Q* to an external circuit.

Which statement is correct?

A The energy dissipation in the external circuit is *EQ*.

B The energy dissipation within the cell is *EQ*.

C The external resistance is *EQ*.

D The total energy dissipation in the cell and the external circuit is *EQ*.

140 **9702/13/M/J/15/Q32**

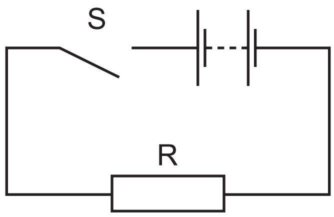
A pedal bicycle is fitted with an electric motor. The rider switches on the motor for a time of 3.0 minutes. A constant current of 3.5 A in the electric motor is provided from a battery with a terminal voltage of 24 V.

What is the energy supplied by the battery?

A 84 J **B** 250 J **C** 630 J **D** 15 000 J

141 9702/13/M/J/15/Q33

The diagram shows a simple circuit.



Which statement is correct?

A When switch S is closed, the electromotive force (e.m.f.) of the battery falls because work is done against the internal resistance of the battery.

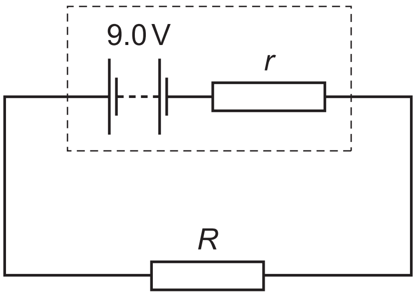
B When switch S is closed, the e.m.f. of the battery falls because work is done against the resistance of R.

C When switch S is closed, the potential difference across the battery falls because work is done against the internal resistance of the battery.

D When switch S is closed, the potential difference across the battery falls because work is done against the resistance of R.

142 9702/13/M/J/15/Q34

A simple circuit is formed by connecting a resistor of resistance R between the terminals of a battery of electromotive force (e.m.f.) 9.0 V and constant internal resistance r.



A charge of 6.0 C flows through the resistor in a time of 2.0 minutes causing it to dissipate 48 J of thermal energy.

What is the internal resistance r of the battery?

A 0.17 Ω **B** 0.33 Ω **C** 20 Ω **D** 160 Ω

143 9702/13/M/J/15/Q35

A source of e.m.f. 9.0 mV has an internal resistance of 6.0 Ω.

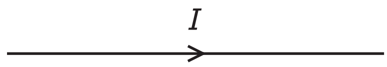
It is connected across a galvanometer of resistance 30 Ω.

What is the current in the galvanometer?

A 250 μA **B** 300 μA **C** 1.5 mA **D** 2.5 mA

144 **9702/11/M/J/17/Q32**

The diagram shows the symbol for a wire carrying a current I.



What does this current represent?

A the charge flowing past a point in the wire per unit time

B the number of electrons flowing past a point in the wire per unit time

C the number of positive ions flowing past a point in the wire per unit time

D the number of protons flowing past a point in the wire per unit time

145 **9702/11/M/J/17/Q33**

Which values of current and resistance will produce a rate of energy transfer of $16\,J\,s^{-1}$?

	current / A	resistance / Ω
A	1	4
B	2	8
C	4	1
D	16	1

146 **9702/11/M/J/17/Q34**

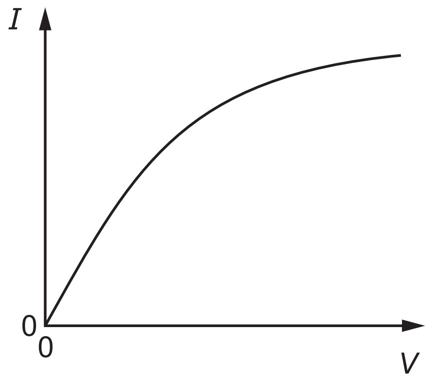
Which component has the I-V graph shown?

A filament lamp

B metallic conductor at constant temperature

C resistor of fixed resistance

D semiconductor diode



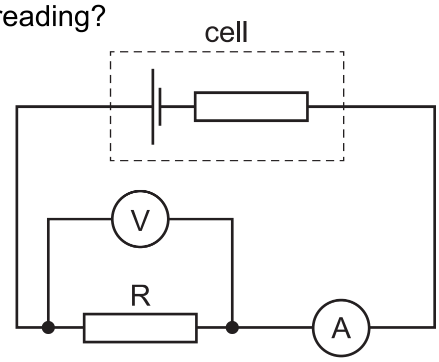
147 **9702/11/M/J/17/Q35**

The circuit shown includes a cell of constant internal resistance and an external resistor R.

A student records the ammeter and voltmeter readings. She then connects a second identical external resistor in parallel with the first external resistor.

What happens to the ammeter reading and to the voltmeter reading?

	ammeter reading	voltmeter reading
A	decreases	decreases
B	decreases	stays the same
C	increases	decreases
D	increases	stays the same



148 **9702/12/M/J/17/Q32**

The current in a circuit component is 2.00 μA.

How many electrons pass through the component each second?

A 1.25×10^{13} **B** 1.25×10^{16} **C** 1.25×10^{19} **D** 1.25×10^{25}

149 **9702/12/M/J/17/Q33**

The filament of a 240 V, 100 W electric lamp heats up from room temperature to its operating temperature. As it heats up, its resistance increases by a factor of 16.

What is the resistance of the filament at room temperature?

A 36 Ω **B** 580 Ω **C** 1.5 kΩ **D** 9.2 kΩ

150 **9702/12/M/J/17/Q32***

Two wires have the same length and the same resistance. Wire X is made of a metal of resistivity $1.7 \times 10^{-8}\,\Omega$ m and wire Y is made of a metal of resistivity $5.6 \times 10^{-8}\,\Omega$ m.

The diameter of wire X is 0.315 mm.

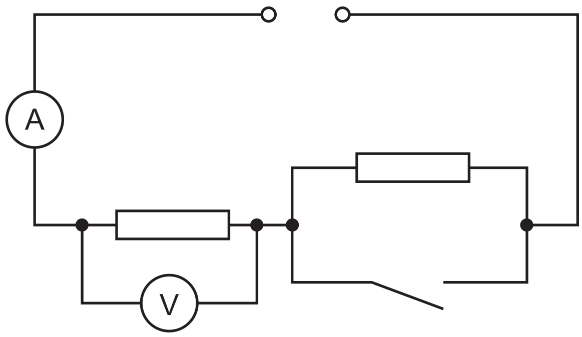
What is the diameter of wire Y?

A 0.17 mm **B** 0.33 mm **C** 0.57 mm **D** 1.0 mm

151 **9702/11/O/N/17/Q36**

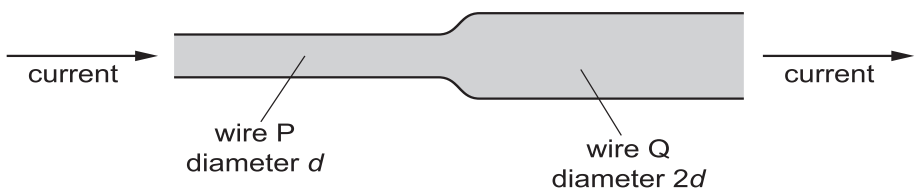
In the circuit shown, the ammeter reading is I and the voltmeter reading is V.
When the switch is closed, which row describes what happens to I and to V?

	I	V
A	decreases	decreases
B	increases	increases
C	increases	stays the same
D	stays the same	increases



152 **9702/12/O/N/17/Q33**

Two copper wires are joined together and carry a current, as shown.



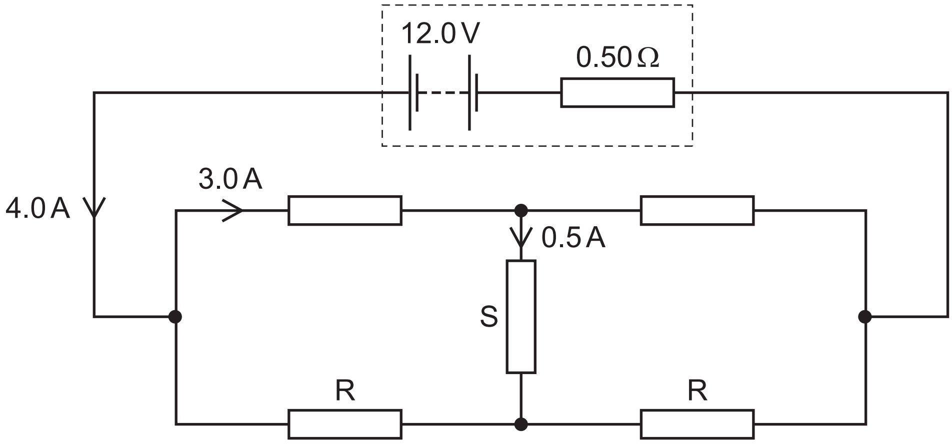
Wire P has diameter d and wire Q has diameter $2d$.

What is the ratio $\dfrac{\text{average drift speed of the free electrons in wire P}}{\text{average drift speed of the free electrons in wire Q}}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** 2 **D** 4

153 **9702/12/O/N/17/Q37**

The circuit shown contains a resistor S that is neither in series nor in parallel with the other resistors.



Kirchhoff's laws can be used with the data in the diagram to deduce the resistance of each of the two identical resistors labelled R. What is the resistance of each resistor R?

A 3.0Ω **B** 4.0Ω **C** 4.8Ω **D** 5.0Ω

154 **9702/13/O/N/17/Q33**

The number density of conduction electrons in copper is $8.0 \times 10^{28}\,m^{-3}$.

What is the average drift speed of electrons in a copper wire of diameter 0.42mm when the current in the wire is 0.57A?

A $8.0 \times 10^{-11}\,m\,s^{-1}$

B $3.2 \times 10^{-10}\,m\,s^{-1}$

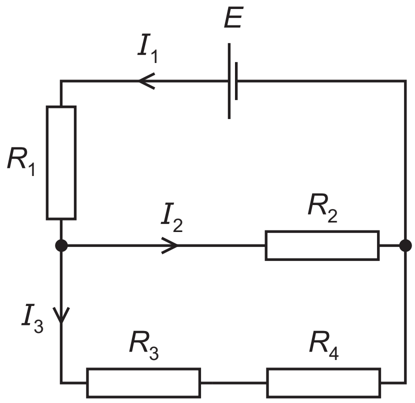
C $8.0 \times 10^{-5}\,m\,s^{-1}$

D $3.2 \times 10^{-4}\,m\,s^{-1}$

155 **9702/11/O/N/17/Q37**

A cell of electromotive force E and negligible internal resistance is connected to a network of resistors of resistances R_1, R_2, R_3 and R_4 as shown.

The branches of the circuit have currents I_1, I_2 and I_3.



Which equation is correct?

A $I_1R_1 + I_2R_2 = I_3R_3 + I_3R_4$

B $I_2R_2 - I_3R_4 - I_3R_3 = 0$

C $E = I_1R_1 + I_2R_2 + I_3R_3 + I_3R_4$

D $E = I_1R_1 + I_2R_2 - I_3R_3 - I_3R_4$

156 **9702/13/O/N/17/Q34**

A simple circuit comprises a source of electromotive force (e.m.f.) connected to a load.

How does the output power P of the source depend on the internal resistance r of the source and the resistance R of the load?

A P is independent of both r and R.

B P depends on r but not on R.

C P depends on R but not on r.

D P depends on both r and R.

157 **9702/11/M/J/18/Q30**

The current I in a metal wire is given by the expression shown. $I = Anvq$

What does the symbol n represent?

A the number of atoms per unit volume of the metal

B the number of free electrons per atom in the metal

C the number of free electrons per unit volume of the metal

D the total number of electrons per unit volume of the metal

158 **9702/11/M/J/18/Q32**

A cylindrical piece of a soft, electrically-conducting material has resistance R. It is rolled out so that its length is doubled but its volume stays constant.

What is its new resistance?

A $\frac{R}{2}$ **B** R **C** $2R$ **D** $4R$

159 **9702/11/M/J/18/Q33**

The sum of the electrical currents into a point in a circuit is equal to the sum of the currents out of the point.

Which statement is correct?

A This is Kirchhoff's first law, which results from the conservation of charge.

B This is Kirchhoff's first law, which results from the conservation of energy.

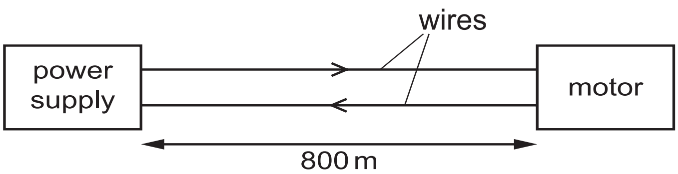
C This is Kirchhoff's second law, which results from the conservation of charge.

D This is Kirchhoff's second law, which results from the conservation of energy.

160 **9702/11/M/J/18/Q37**

A motor is required to operate at a distance of 800 m from its power supply. The motor requires a potential difference (p.d.) of 16.0 V and a current of 0.60 A to operate.

Two wires are used to supply power to the motor as shown.



The resistance of each of these wires is 0.0050 Ω per metre.

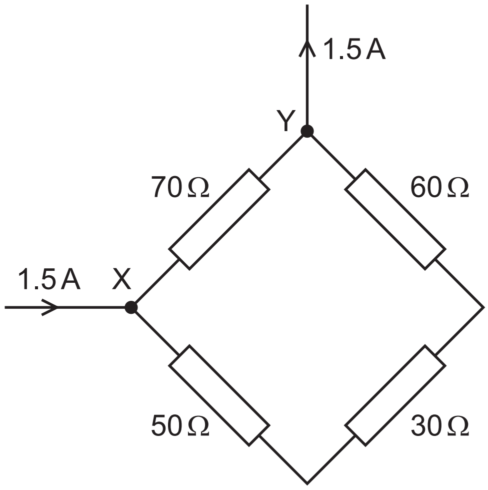
What is the minimum output p.d. of the power supply?

A 11.2 V **B** 16.0 V **C** 18.4 V **D** 20.8 V

161 **9702/12/M/J/18/Q34**

Four different resistors are arranged as shown.

A current of 1.5 A enters the network at junction X and leaves through junction Y.

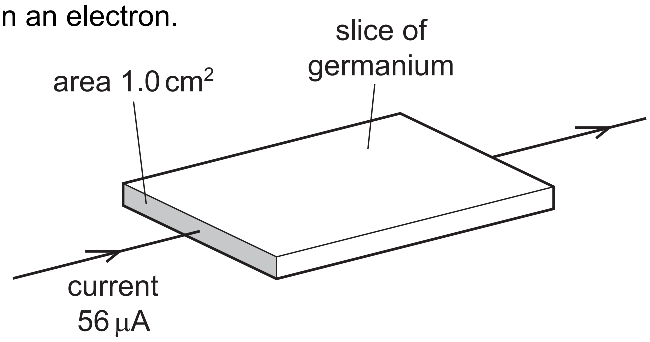


What is the current in the resistor of resistance 30 Ω?

A 0.21 A **B** 0.50 A **C** 0.75 A **D** 1.0 A

162 **9702/13/M/J/18/Q30**

A slice of germanium of cross-sectional area $1.0\,cm^2$ carries a current of $56\,\mu A$. The number density of charge carriers in the germanium is $2.0 \times 10^{13}\,cm^{-3}$. Each charge carrier has a charge equal to the charge on an electron.



What is the average drift velocity of the charge carriers in the germanium?

A $0.18\,m\,s^{-1}$ **B** $18\,m\,s^{-1}$ **C** $180\,m\,s^{-1}$ **D** $1800\,m\,s^{-1}$

163 **9702/13/M/J/18/Q33**

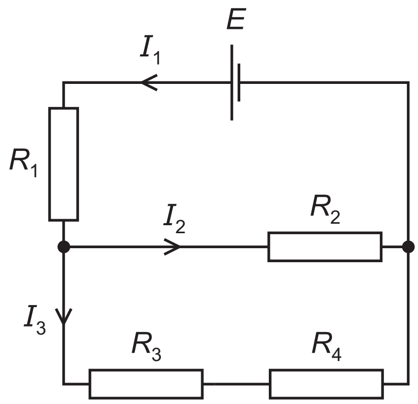
A metal wire of length 1.4 m has a uniform cross-sectional area of $7.8 \times 10^{-7}\,m^2$.

The resistivity of the metal is $1.7 \times 10^{-8}\,\Omega\,m$. What is the resistance of the wire?

A 0.016 Ω **B** 0.031 Ω **C** 33 Ω **D** 64 Ω

164 9702/11/O/N/17/Q37

A cell of electromotive force E and negligible internal resistance is connected to a network of resistors of resistances R_1, R_2, R_3 and R_4 as shown.



The branches of the circuit have currents I_1, I_2 and I_3.

Which equation is correct?

A $I_1R_1 + I_2R_2 = I_3R_3 + I_3R_4$

B $I_2R_2 - I_3R_4 - I_3R_3 = 0$

C $E = I_1R_1 + I_2R_2 + I_3R_3 + I_3R_4$

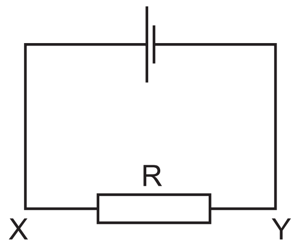
D $E = I_1R_1 + I_2R_2 - I_3R_3 - I_3R_4$

165 9702/11/O/N/17/Q33

The current in the circuit shown is 3.2 mA.

What are the direction of flow and the rate of flow of electrons through the resistor R?

	direction of flow	rate of flow / s^{-1}
A	X to Y	2.0×10^{16}
B	X to Y	5.1×10^{-22}
C	Y to X	2.0×10^{16}
D	Y to X	5.1×10^{-22}



166 9702/12/F/M/18/Q34

A wire of resistance 9.55 Ω has a diameter of 0.280 mm.

It is made of metal of resistivity 4.90×10^{-7} Ω m.

What is the length of the wire?

A 1.20 m **B** 4.80 m **C** 19.0 m **D** 76.8 m

167 9702/12/F/M/18/Q35

Charge carriers, each of charge q, move along a wire of fixed length. The number density of the charge carriers in the wire is n.

What is also required, for this wire, to determine the average drift velocity of the charge carriers in terms of n and q?

A current per unit of cross-sectional area

B potential difference per unit of length

C resistance and cross-sectional area

D resistivity and length

168 **9702/11/O/N/18/Q33**

When there is a current of 5.0 A in a copper wire, the average drift velocity of the free electrons is $8.0 \times 10^{-4}\,\mathrm{m\,s^{-1}}$.

What is the average drift velocity in a different copper wire that has twice the diameter and a current of 10.0 A?

A $4.0 \times 10^{-4}\,\mathrm{m\,s^{-1}}$

B $8.0 \times 10^{-4}\,\mathrm{m\,s^{-1}}$

C $1.6 \times 10^{-3}\,\mathrm{m\,s^{-1}}$

D $3.2 \times 10^{-3}\,\mathrm{m\,s^{-1}}$

169 **9702/11/O/N/18/Q34**

What is equivalent to one volt?

A one coulomb per second

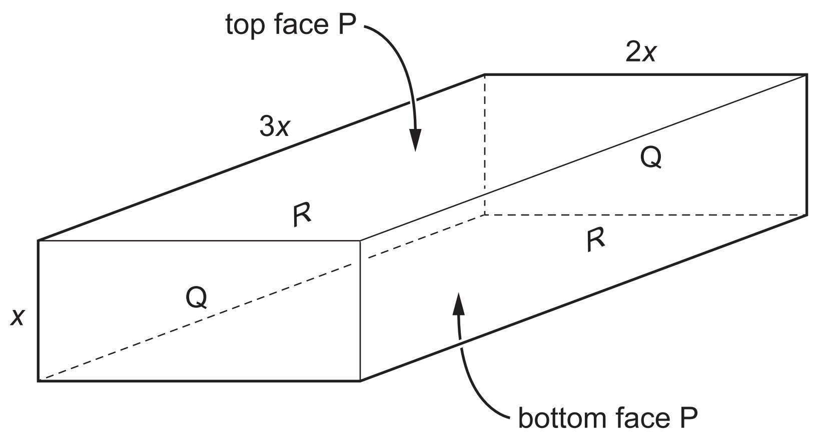
B one joule per coulomb

C one joule per second

D one joule second per coulomb squared

170 **9702/11/O/N/18/Q35**

The diagram shows a rectangular block with dimensions x, $2x$ and $3x$.



Electrical contact can be made to the block between opposite pairs of faces (for example, between the faces labelled R).

Which statement describing the electrical resistance of the block is correct?

A It is maximum between the faces labelled P.

B It is maximum between the faces labelled Q.

C It is maximum between the faces labelled R.

D It is the same, whichever pair of faces is used.

171 **9702/12/O/N/18/Q33**

Which two units are used to define the coulomb?

A ampere and second

B ampere and volt

C volt and ohm

D volt and second

172 **9702/12/O/N/18/Q35**

A wire of length *L* has resistance *R*. The cross-section of the wire is circular with radius *r*.

A second wire, also of circular cross-section, and of the same material, has resistance $\frac{1}{2}R$.

What could be the radius and the length of the second wire?

	radius	length
A	$\frac{r}{2}$	$\frac{L}{2}$
B	$\frac{r}{\sqrt{2}}$	$\frac{L}{2}$
C	$r\sqrt{2}$	$2L$
D	$2r$	$2L$

173 **9702/13/O/N/18/Q32**

The current I in a copper wire can be calculated using the equation shown.

$$I = Anvq$$

What does the symbol v represent?

A the average drift velocity of the charge carriers

B the instantaneous velocity of the charge carriers

C the voltage applied across the wire

D the volume of the wire

174 **9702/13/O/N/18/Q34**

Gold is sometimes used to make very small connecting wires in electronic circuits.

A particular gold wire has length 2.50×10^{-3} m and cross-sectional area 6.25×10^{-8} m^2. Gold has resistivity 2.30×10^{-8} Ω m.

What is the resistance of the wire?

A 3.6×10^{-18} Ω

B 5.8×10^{-13} Ω

C 9.2×10^{-4} Ω

D 6.8×10^{-3} Ω

175 **9702/11/M/J/19/Q32**

In an electrolyte, the electric current is carried by charged particles (ions) in solution.

What is **not** a possible value for the charge on an ion in solution?

A -4.8×10^{-19} C

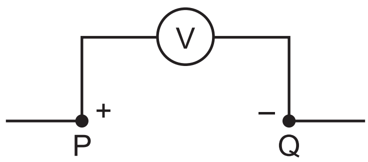
B $+1.6 \times 10^{-19}$ C

C $+3.2 \times 10^{-19}$ C

D $+4.0 \times 10^{-19}$ C

176 **9702/11/M/J/19/Q33**

A voltmeter connected between two points P and Q in an electrical circuit shows a reading of 1 V.



Which statement is correct?

A The energy needed to move +1 C of charge from P to Q is 1 J.

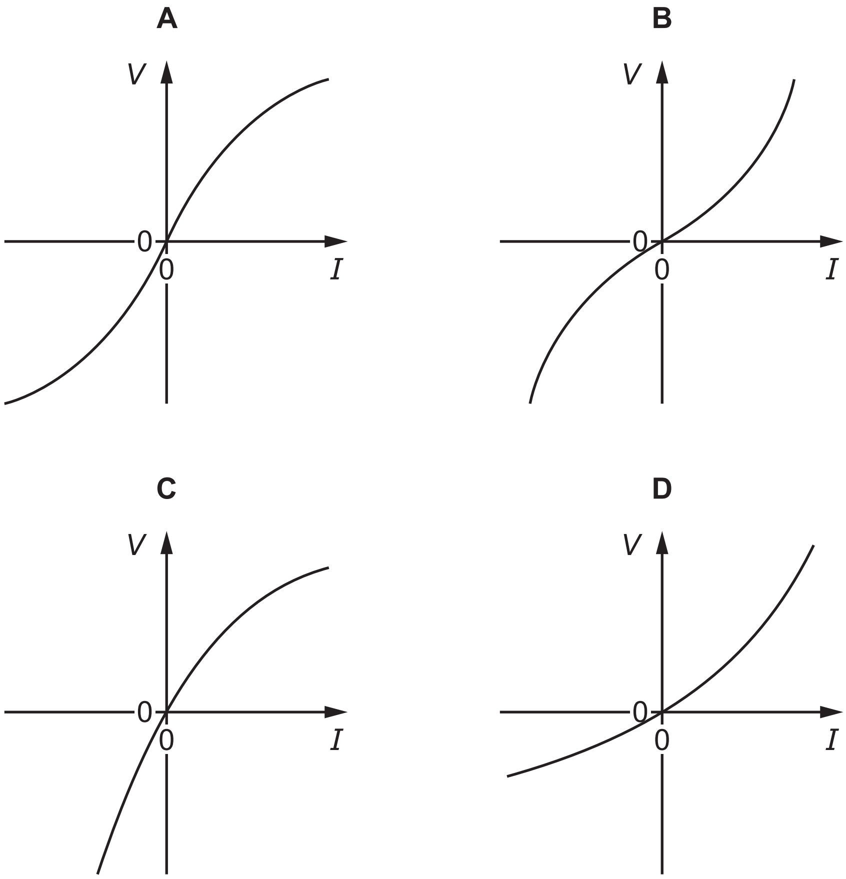
B The energy needed to move +1 C of charge from Q to P is 1 J.

C The energy needed to move one electron from P to Q is 1 J.

D The energy needed to move one electron from Q to P is 1 J.

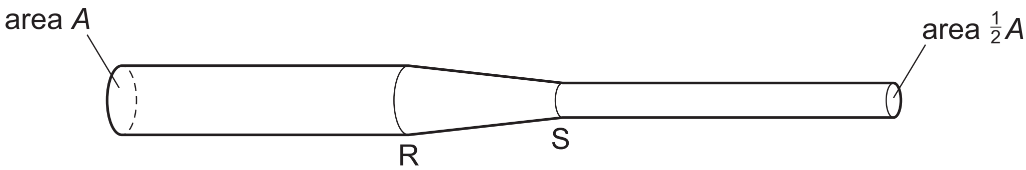
177 9702/11/M/J/19/Q34

Which graph best represents the variation with current I of potential difference V for a filament lamp?



178 9702/12/M/J/19/Q33

A length of wire is connected into a circuit.



The area of the cross-section of the wire changes from A at R to $\frac{1}{2}A$ at S.

There is a constant current in the wire. Charge Q passes R in time t.

What is the charge passing point S in the same time t?

A $\frac{1}{2}Q$ **B** Q **C** $Q\sqrt{2}$ **D** $2Q$

179 9702/12/M/J/19/Q34

Four wires are made of the same metal. The cross-sectional areas, lengths and thermodynamic temperatures of the wires are shown.

Which wire has the largest resistance?

	cross-sectional area	length	temperature
A	A	$2L$	$2T$
B	A	L	T
C	$2A$	$2L$	$2T$
D	$2A$	L	T

180 9702/13/M/J/19/Q32

Which two units are used to define the volt?

A ampere and ohm

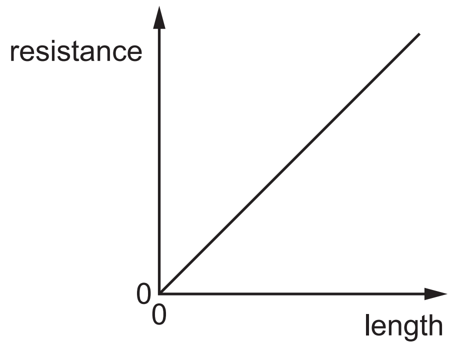
B coulomb and joule

C coulomb and ohm

D coulomb and second

181 9702/13/M/J/19/Q33

The graph shows the variation with length of the resistance of a uniform metal wire.



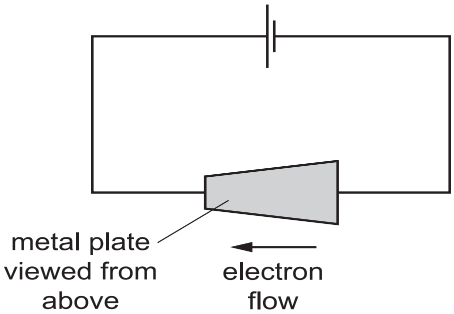
The gradient of the graph is G.
The wire has cross-sectional area A.

Which expression could be used to calculate the resistivity of the metal of the wire?

A $G \times A$ **B** $\frac{G}{A}$ **C** $\frac{A}{G}$ **D** $G \times A^2$

182 9702/11/O/N/19/Q30

A metal plate of uniform thickness is connected to a cell as shown.



Electrons move clockwise around the circuit.

Which statement about the metal plate is correct?

A The average drift speed of the conduction electrons decreases as they move from right to left through the plate.

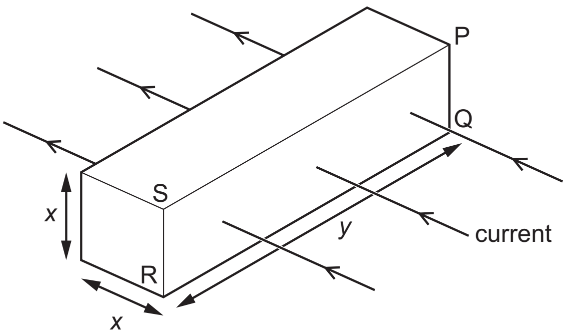
B The average drift speed of the conduction electrons increases as they move from right to left through the plate.

C The number density of the conduction electrons decreases from right to left through the plate.

D The number density of the conduction electrons increases from right to left through the plate.

183 9702/11/O/N/19/Q31

The diagram shows the direction of the current in a metal block. The charge carriers enter the block through the face PQRS and leave the block through the opposite face.



The number density of charge carriers is n. Each charge carrier has charge e. The average drift speed of the charge carriers is v.

Which expression gives the current in the block?

A $envx^2$ **B** $envxy$ **C** $envx^3y^2$ **D** $envx^4y$

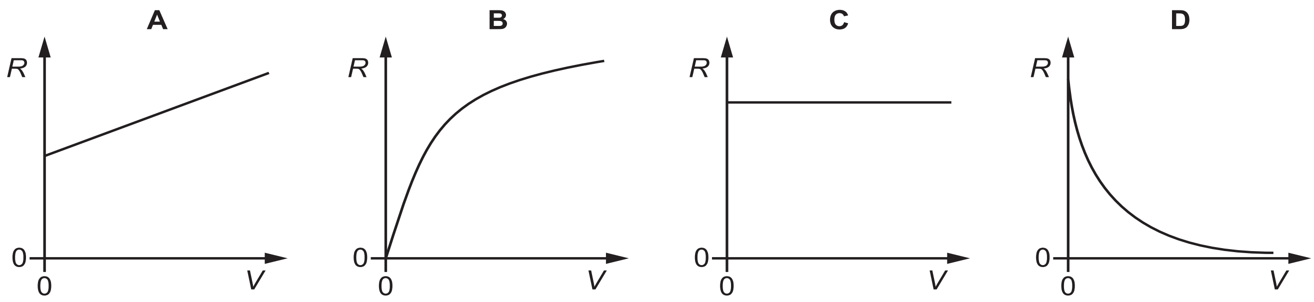
184 9702/11/O/N/19/Q32

What could **not** be used as a unit of potential difference?

A $\mathrm{A\,\Omega}$ **B** $\mathrm{N\,m^{-1}\,C^{-1}}$ **C** $\mathrm{W\,A^{-1}}$ **D** $(\mathrm{\Omega\,W})^{\frac{1}{2}}$

185 9702/11/O/N/19/Q33

Which graph could show how the resistance R of a filament lamp varies with the applied potential difference (p.d.) V, as V is increased to the normal operating p.d.?



186 9702/12/O/N/19/Q31

When the current in a wire is 5.0 A, the average drift speed of the conduction electrons in the wire is $7.4 \times 10^{-4}\,\mathrm{m\,s^{-1}}$.

Which row gives a possible cross-sectional area and number of conduction electrons per unit volume for this wire?

	cross-sectional area / m^2	number of conduction electrons per unit volume / m^{-3}
A	7.2×10^{-7}	1.2×10^{28}
B	7.2×10^{-7}	5.9×10^{28}
C	2.3×10^{-6}	7.3×10^{26}
D	2.3×10^{-6}	3.7×10^{27}

187 9702/12/O/N/19/Q32

A fixed resistor of resistance 12 Ω is connected to a battery. There is a current of 0.20 A in the resistor. The current is now doubled.

What is the new power dissipated in the resistor?

A 0.48 W **B** 0.96 W **C** 1.9 W **D** 4.8 W

188 9702/12/O/N/19/Q33

There is a current in a resistor for an unknown time.

Which two quantities can be used to calculate the energy dissipated by the resistor?

A the current in the resistor and the potential difference across the resistor

B the resistance of the resistor and the current in the resistor

C the total charge passing through the resistor and the potential difference across the resistor

D the total charge passing through the resistor and the resistance of the resistor

189 9702/13/O/N/19/Q33

A metal electrical conductor has a resistance of 5.6 kΩ. A potential difference (p.d.) of 9.0 V is applied across its ends.

How many electrons pass a point in the conductor in one minute?

A 6.0×10^{20} **B** 1.0×10^{19} **C** 6.0×10^{17} **D** 1.0×10^{16}

190. 9702/12/F/M/20/Q32

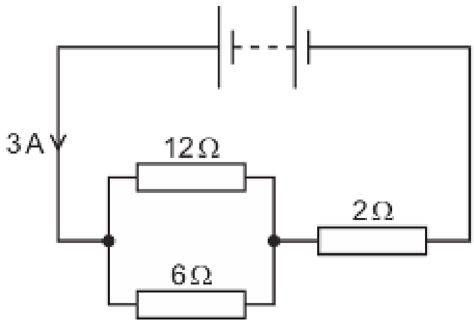
Electrons move in a vacuum from one metal plate to another metal plate. As a result of this, there is an electric current of 48 μA between the two plates.

How many electrons are emitted by the first plate in a time of 5.0 minutes?

A 1.4×10^{4} **B** 1.5×10^{15} **C** 1.8×10^{16} **D** 9.0×10^{16}

191. 9702/12/F/M/20/Q33

A battery is connected to three resistors of resistances 12 Ω, 6 Ω and 2 Ω, as shown.



The current from the battery is 3 A.

What is the value of the ratio $\dfrac{\text{power dissipated in the resistor of resistance } 6\,\Omega}{\text{power dissipated in the resistor of resistance } 2\,\Omega}$?

A $\frac{1}{3}$ **B** $\frac{4}{3}$ **C** $\frac{2}{1}$ **D** $\frac{3}{1}$

192. 9702/12/F/M/20/Q34

A manufacturer recommends that the longer the extension cord you use with an electric drill, the bigger the cross-sectional area of the cord should be.

What is a reason for this recommendation?

A Resistance is inversely proportional to both the length and the cross-sectional area.

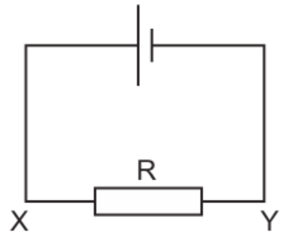
B Resistance is inversely proportional to the length and directly proportional to the cross-sectional area.

C Resistance is proportional to both the length and the cross-sectional area.

D Resistance is proportional to the length and inversely proportional to the cross-sectional area.

193. 9702/11/M/J/20/Q33

The current in the circuit shown is 3.2 mA.

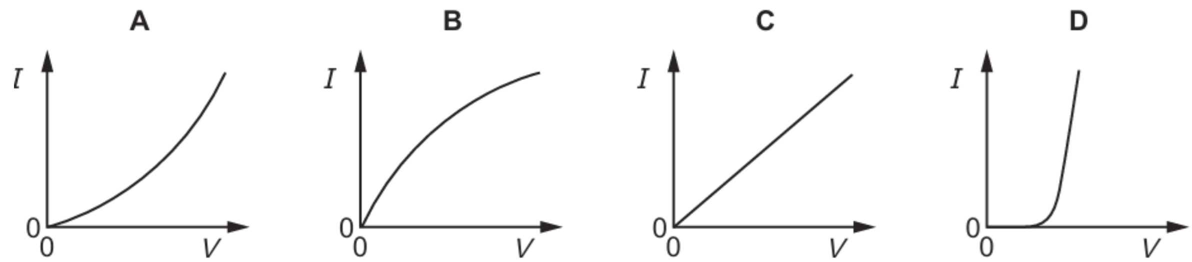


What are the direction of flow and the rate of flow of electrons through the resistor R?

	direction of flow	rate of flow / s^{-1}
A	X to Y	2.0×10^{16}
B	X to Y	5.1×10^{-22}
C	Y to X	2.0×10^{16}
D	Y to X	5.1×10^{-22}

194. 9702/11/M/J/20/Q34

Which graph best represents the way the current I through a filament lamp varies with the potential difference V across it?



195. 9702/11/M/J/20/Q35

A cylindrical metal wire X has resistance R. The same volume of the same metal is made into a cylindrical wire Y of double the length.

What is the resistance of wire Y?

A R **B** $2R$ **C** $4R$ **D** $8R$

196. 9702/12/M/J/20/Q34

An electric kettle is rated at 2.0 kW, which describes the power supplied to the heating coil in the kettle.

The coil has a resistance of 5.0 kΩ.

What is the current in the coil?

A 0.40 A **B** 0.63 A **C** 1.6 A **D** 2.5 A

197. 9702/12/M/J/20/Q33

The number density of free electrons in copper is $8.0 \times 10^{28}\,m^{-3}$.

A copper wire has diameter 0.42 mm.

What is the average drift speed of the free electrons in the wire when the current in the wire is 0.57 A?

A $8.0 \times 10^{-11}\,m\,s^{-1}$

B $3.2 \times 10^{-10}\,m\,s^{-1}$

C $8.0 \times 10^{-5}\,m\,s^{-1}$

D $3.2 \times 10^{-4}\,m\,s^{-1}$

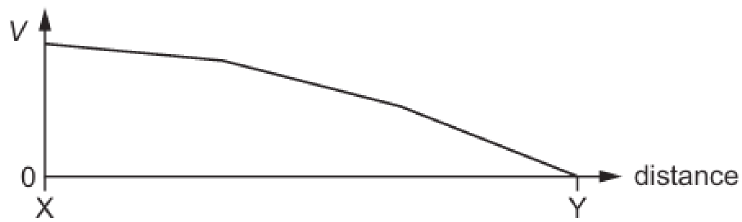
198. 9702/12/M/J/20/Q35

A conductor consists of three wires connected in series. The wires are all made of the same metal but have different cross-sectional areas. There is a current I in the conductor.

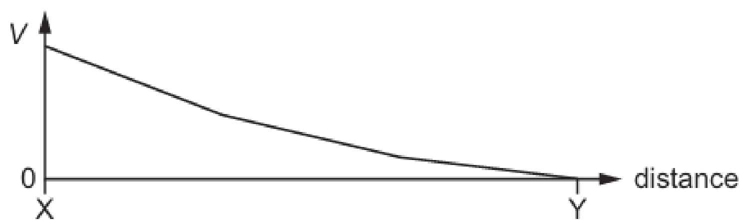
Point Y on the conductor is at zero potential.

Which graph best shows the variation of potential V with distance along the conductor?

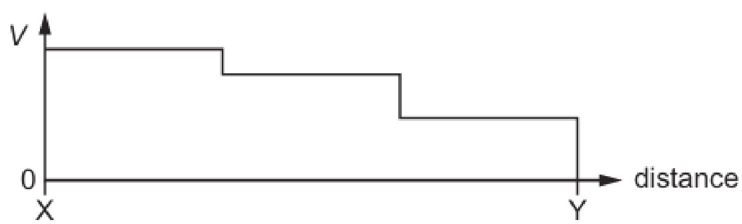
A



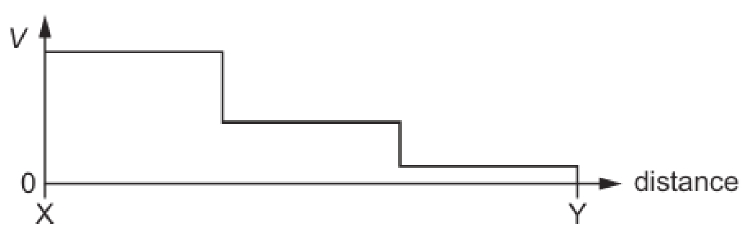
B



C



D



199. 9702/13/M/J/20/Q33

The unit of electric charge is the coulomb.

What is meant by 1 coulomb?

A the charge passing a point in 1 second when a current produces 1 joule of work

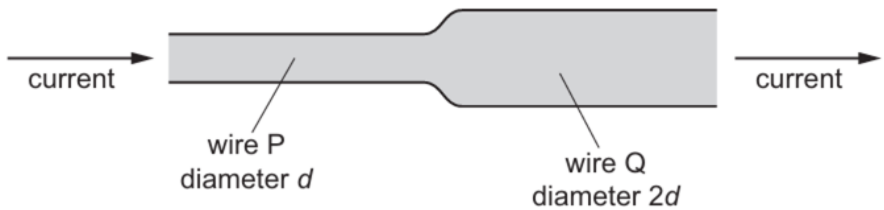
B the charge passing a point in 1 second when a current produces 1 watt of power

C the charge passing a point in 1 second when there is a current of 1 ampere

D the charge passing a point in 1 second when there is 1 ohm of resistance

200. 9702/13/M/J/20/Q34

Two copper wires are joined together and carry a current, as shown.



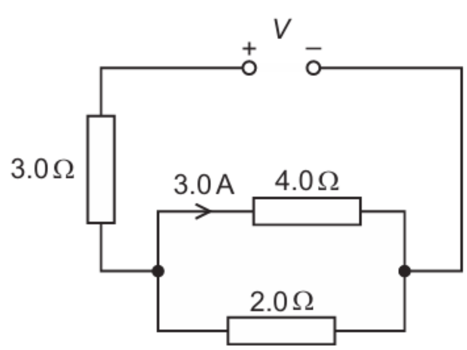
Wire P has diameter d and wire Q has diameter $2d$.

What is the ratio $\dfrac{\text{average drift speed of the free electrons in wire P}}{\text{average drift speed of the free electrons in wire Q}}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** 2 **D** 4

201. 9702/13/M/J/20/Q35

A power supply of electromotive force (e.m.f.) V and negligible internal resistance is connected in the circuit shown. There is a current of 3.0 A in the 4.0 Ω resistor.



What is the value of V?

A 15 V **B** 29 V **C** 39 V **D** 51 V

202. 9702/13/M/J/20/Q36

The wire of a heating element has resistance R. The wire breaks and is replaced by a different wire.

Data for the original wire and for the replacement wire are shown in the table.

	length	diameter	resistivity of metal
original wire	l	d	ρ
replacement wire	l	$2d$	2ρ

What is the resistance of the replacement wire?

A $\dfrac{R}{4}$ **B** $\dfrac{R}{2}$ **C** R **D** $2R$

203. 9702/11/O/N/20/Q32

Free electrons flow along a copper wire X of radius 5.0×10^{-5} m with an average drift speed of $2.8 \times 10^{-2}\,\text{m}\,\text{s}^{-1}$. The current in the wire is 3.0 A.

There is a current of 2.0 A in a copper wire Y of radius 1.0×10^{-4} m.

What is the average drift speed of the free electrons in copper wire Y?

A $4.7 \times 10^{-3}\,\text{m}\,\text{s}^{-1}$

B $9.3 \times 10^{-3}\,\text{m}\,\text{s}^{-1}$

C $1.1 \times 10^{-2}\,\text{m}\,\text{s}^{-1}$

D $1.9 \times 10^{-2}\,\text{m}\,\text{s}^{-1}$

204. 9702/11/O/N/20/Q33

What is the definition of potential difference?

A power per unit current

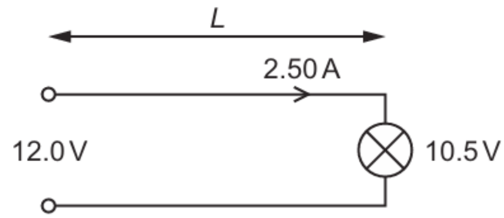
B product of current and resistance

C product of electric field strength and distance

D work done per unit charge

205. 9702/11/O/N/20/Q34

A cable of length L consisting of two wires is used to connect a 12.0 V power supply of negligible internal resistance to a lamp, as shown.



The potential difference across the lamp is 10.5 V. The current in the wire is 2.50 A.

Each wire is made of metal of resistivity $1.70 \times 10^{-8}\,\Omega\,m$ and has a cross-sectional area of $6.00 \times 10^{-7}\,m^2$.

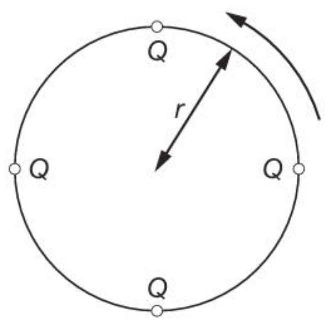
What is the length L of the cable?

A 10.6 m **B** 21.2 m **C** 29.4 m **D** 58.8 m

206. 9702/12/O/N/20/Q33

Four point charges, each of charge Q, are placed on the edge of an insulating disc of radius r.

The disc rotates at a rate of n revolutions per unit time.

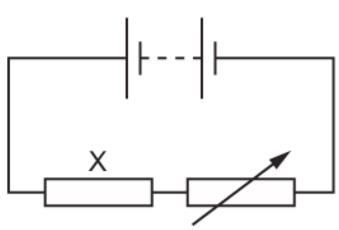


What is the equivalent electric current at the edge of the disc?

A $4Qn$ **B** $\frac{4Q}{n}$ **C** $8\pi rQn$ **D** $\frac{2Qn}{\pi r}$

207. 9702/12/O/N/20/Q34

In the circuit shown, a fixed resistor X is connected in series with a battery and a variable resistor.



The power dissipated in resistor X is 7.2 W when a current of 3.0 A passes through it.

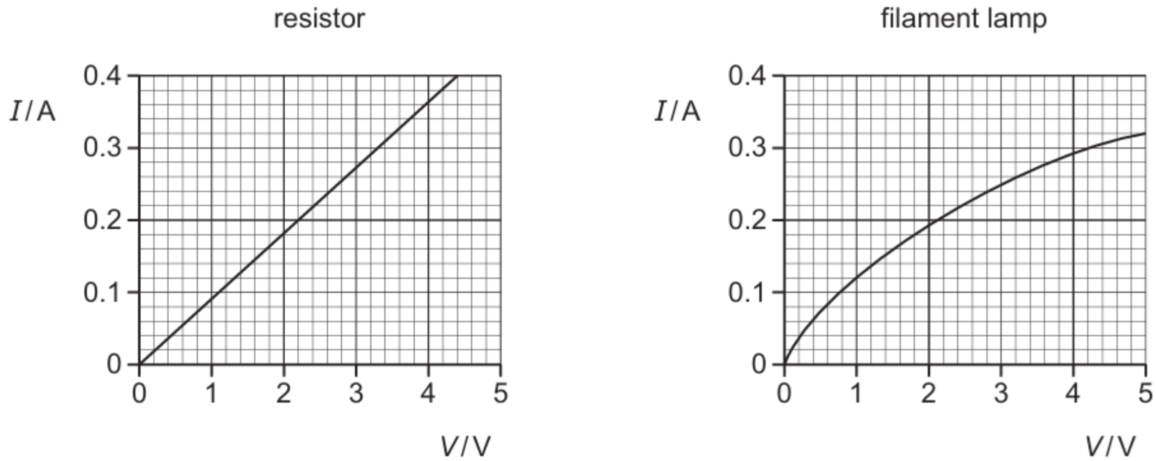
The variable resistor is adjusted so that the power dissipated in X increases by 50%.

What is the new current in the circuit?

A 2.4 A **B** 3.7 A **C** 4.5 A **D** 14 A

208. 9702/12/O/N/20/Q35

A resistor and a filament lamp are connected in series with a power supply. The I–V characteristics of the resistor and of the lamp are shown below.



The potential difference (p.d.) across the resistor is 3.3 V.

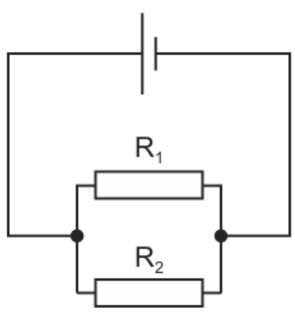
What is the resistance of the lamp?

A 0.071 Ω **B** 4.2 Ω **C** 11 Ω **D** 14 Ω

209. 9702/13/O/N/20/Q33

Two resistors R_1 and R_2 are made from wire of the same material.

They are connected in parallel to each other in a circuit, as shown.



The diameter of R_2 is half the diameter of R_1.

The resistance of R_2 is three times the resistance of R_1.

What is the value of the ratio $\frac{\text{average drift speed of free electrons in } R_1}{\text{average drift speed of free electrons in } R_2}$?

A $\frac{3}{2}$ **B** $\frac{3}{4}$ **C** $\frac{1}{6}$ **D** $\frac{1}{12}$

210. 9702/13/O/N/20/Q34

A student describes potential difference as the energy transferred per unit charge.

Which statement about the energy transfer is correct?

A It is from electrical energy into other forms.

B It is from other forms into electrical energy.

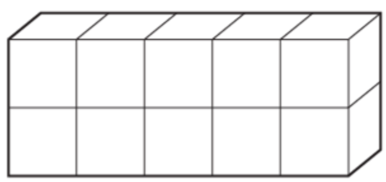
C It only takes place inside a power supply.

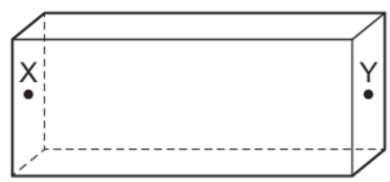
D It only takes place inside resistors.

211. 9702/13/O/N/20/Q35

A metal cube has a resistance of $4.0\,\Omega$ between opposite faces.

Ten of these cubes are put together to make a cuboid of $1 \times 2 \times 5$ cubes.





There is no extra resistance where the faces of the cubes touch each other.

What is the resistance of the cuboid when connected between faces X and Y?

A $1.6\,\Omega$ **B** $2.0\,\Omega$ **C** $10\,\Omega$ **D** $40\,\Omega$

212. 9702/12/F/M/21/Q32

The current I in a metal wire is given by the equation

$$I = Anvq.$$

What does the symbol n represent?

A the number of charge carriers in the wire

B the number of charge carriers per unit cross-sectional area of the wire

C the number of charge carriers per unit length of the wire

D the number of charge carriers per unit volume of the wire

213. 9702/12/F/M/21/Q35

An electrical cable consists of seven strands of copper wire, each of diameter 0.30 mm, connected in parallel.

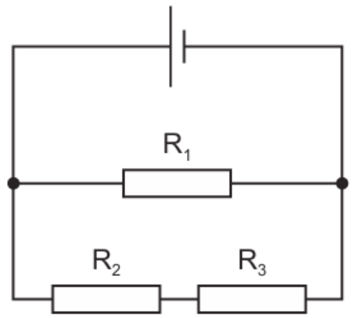
The resistivity of copper is $1.72 \times 10^{-8}\,\Omega\,\text{m}$. The current in the cable is 13 A.

What is the potential difference (p.d.) between two points on the cable a distance of 1.0 m apart?

A 0.0045 V **B** 0.11 V **C** 0.45 V **D** 3.2 V

214. 9702/12/F/M/21/Q33

A cell of negligible internal resistance is connected to resistors R_1, R_2 and R_3, as shown. The cell provides power to the circuit and power is dissipated in the resistors.



Which word equation **must** be correct?

A power dissipated in R_1 = power dissipated in R_2 + power dissipated in R_3

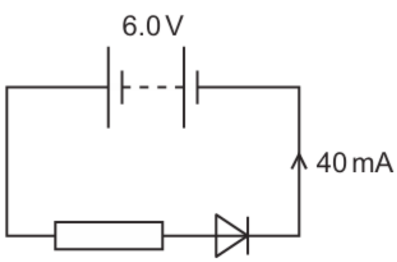
B power dissipated in R_2 = power dissipated in R_3

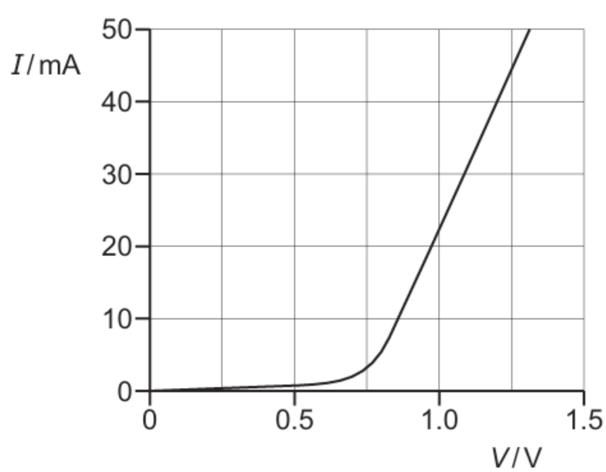
C power output of cell = power dissipated in R_1 + power dissipated in R_2 + power dissipated in R_3

D power output of cell = power dissipated in R_1

215. 9702/12/F/M/21/Q34

A fixed resistor and a diode are connected in series to a battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance. The graph shows the variation with potential difference (p.d.) *V* of the current *I* for the diode.





The current in the diode is 40 mA.

What is the resistance of the fixed resistor?

A 30 Ω **B** 120 Ω **C** 150 Ω **D** 180 Ω

216. 9702/11/M/J/21/Q32
Which two units are used to define the coulomb?

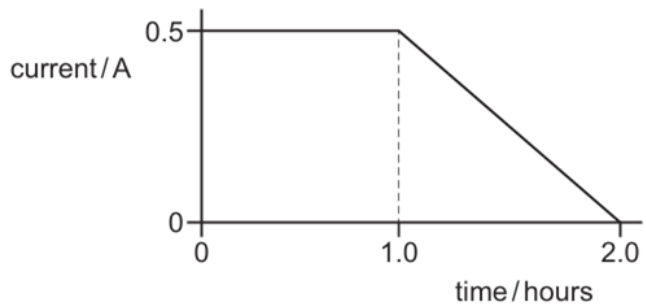
A ampere and second

B ampere and volt

C volt and ohm

D volt and second

217. 9702/11/M/J/21/Q33
A mobile phone battery is charged by connecting it to a constant potential difference of 5.0 V. After a time of 1.0 hour, the initial current of 0.50 A slowly decreases to zero, as shown.



What is the best estimate of the energy transferred to the battery during the time of 2.0 hours shown in the graph?

A 2700 J **B** 9000 J **C** 14 000 J **D** 18 000 J

218. 9702/11/M/J/21/Q34
A length of wire is connected into an electric circuit. The current in the wire is measured.

Which change on its own could increase the current in the wire?

A an increase in the length of the wire

B an increase in the radius of the wire

C an increase in the resistance of the wire

D an increase in the resistivity of the wire

219. 9702/12/M/J/21/Q35
A wire of resistance 9.55 Ω has a diameter of 0.280 mm.

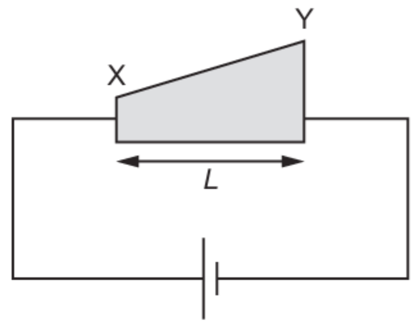
It is made of metal of resistivity $4.90 \times 10^{-7}\,\Omega\,\text{m}$.

What is the length of the wire?

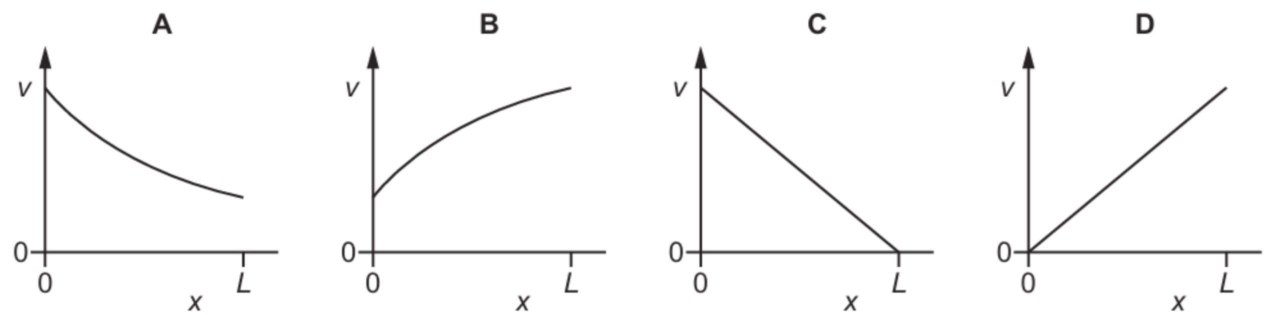
A 1.20 m **B** 4.80 m **C** 19.0 m **D** 76.0 m

220. 9702/12/M/J/21/Q32

A wedge-shaped metal conductor of length L, varying width and uniform thickness is connected to a cell, as shown.

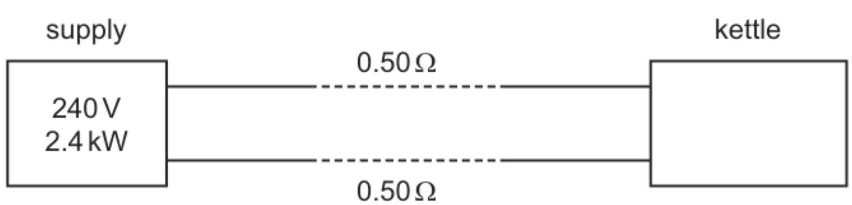


Which graph best shows how the average drift velocity v of electrons in the conductor varies with distance x from end X?



221. 9702/12/M/J/21/Q33

The power output of an electrical supply is 2.4 kW at a potential difference (p.d.) of 240 V. The two wires between the supply and a kettle each have a resistance of 0.50 Ω, as shown.

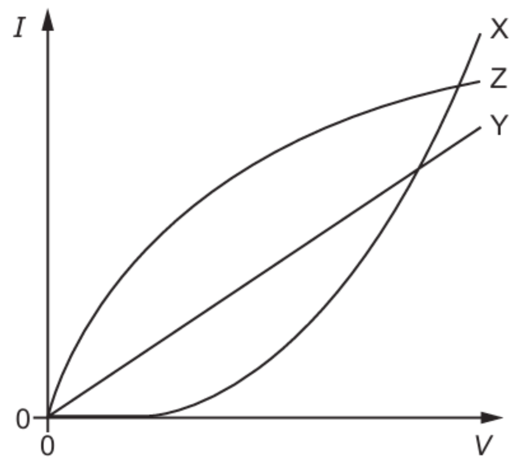


What is the power supplied to the kettle and what is the p.d. across the kettle?

	power / kW	p.d. / V
A	2.3	230
B	2.3	235
C	2.4	230
D	2.4	235

222. 9702/12/M/J/21/Q34

The graph shows the variation with potential difference V of the current I in components X, Y and Z.

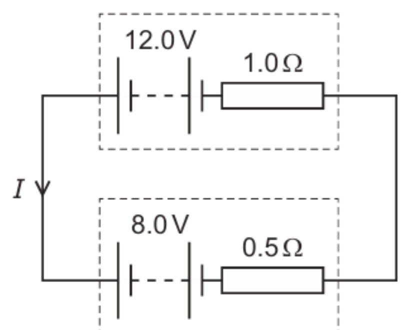


Which row correctly identifies the components?

	metallic conductor at constant temperature	semiconductor diode	filament lamp
A	X	Z	Y
B	Y	X	Z
C	Y	Z	X
D	Z	Y	X

223. 9702/13/M/J/21/Q35

The diagram shows a circuit containing two batteries connected together.



What is the current I?

A 2.7 A **B** 4.0 A **C** 8.0 A **D** 13 A

224. 9702/13/M/J/21/Q34

A battery is marked 9.0 V.

What does this mean?

A Each coulomb of charge from the battery supplies 9.0 J of electrical energy to the whole circuit.

B The battery supplies 9.0 J of electrical energy to an external circuit for each coulomb of charge.

C The potential difference across any component connected to the battery will be 9.0 V.

D There will always be a potential difference of 9.0 V across the battery terminals.

225. 9702/13/M/J/21/Q33

A wire has a length of 12 cm and contains a total of 5.1×10^{22} free electrons.

When a potential difference is applied across the ends of the wire, the free electrons move with an average drift speed of $4.0 \times 10^{-6}\,\mathrm{m\,s^{-1}}$.

What is the current in the wire?

A 0.0027 A **B** 0.0039 A **C** 0.27 A **D** 0.39 A

226. 9702/11/O/N/21/Q32

The current I in a wire is given by the equation

$$I = nAvq$$

where n is the number density of the free electrons, A is the cross-sectional area of the wire, v is the average drift velocity of the free electrons and q is the charge of an electron.

Which relationship is **not** used in the derivation of this equation?

A charge = current × time

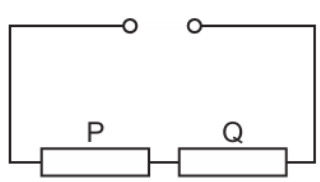
B distance = speed × time

C number = number density × area

D volume = length × area

227. 9702/11/O/N/21/Q33

A circuit contains two resistors, P and Q, and a power supply of negligible internal resistance, as shown.



The current in resistor P is 2.0 A and the power dissipated by resistor P is 18 W.

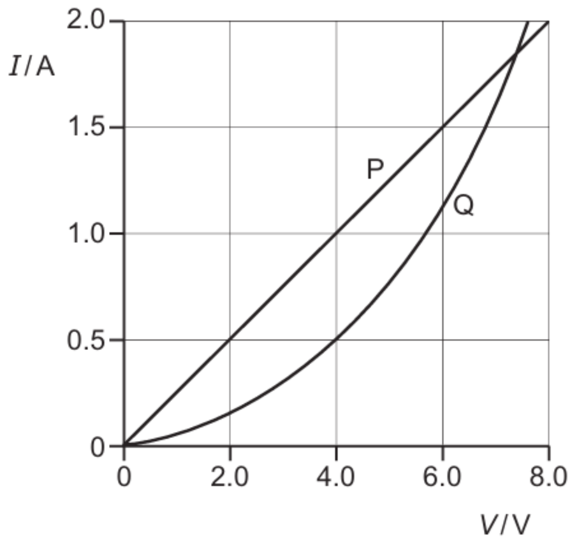
Resistor Q dissipates 240 J of energy when a charge of 40 C passes through it.

What is the electromotive force (e.m.f.) of the power supply?

A 3.0 V **B** 6.0 V **C** 9.0 V **D** 15 V

228. 9702/11/O/N/21/Q34

The I–V characteristics of two electrical components P and Q are shown.



Which statement is correct?

A For a current of 0.5 A, the power dissipated in Q is double that in P.

B For a current of 1.9 A, the resistance of Q is approximately half that of P.

C The resistance of Q increases as the current in it increases.

D P is a fixed resistor and Q is a filament lamp.

229. 9702/11/O/N/21/Q35

Two copper wires S and T, of equal length, are connected in parallel. Wire S has a diameter of 3.0 mm. Wire T has a diameter of 1.5 mm.

A potential difference is applied across the ends of this parallel arrangement.

What is the value of the ratio $\frac{\text{current in S}}{\text{current in T}}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** 2 **D** 4

230. 9702/12/O/N/21/Q33

What is a description of the coulomb?

A the electric charge of one electron

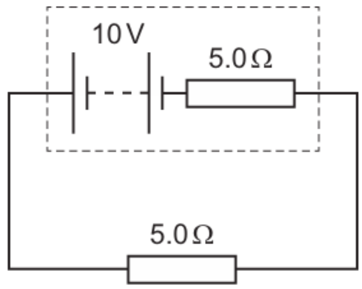
B the electric charge transferred by a current of one ampere in one second

C the kinetic energy gained by an electron accelerated through a potential difference of one volt

D the kinetic energy of an electron moving at a speed of one metre per second

231. 9702/12/O/N/21/Q34

A battery of electromotive force (e.m.f.) 10 V and internal resistance 5.0 Ω is connected to a 5.0 Ω load resistor.



Which change occurs when the 5.0 Ω load resistor is replaced with a 50 Ω load resistor?

A The current in the circuit increases.

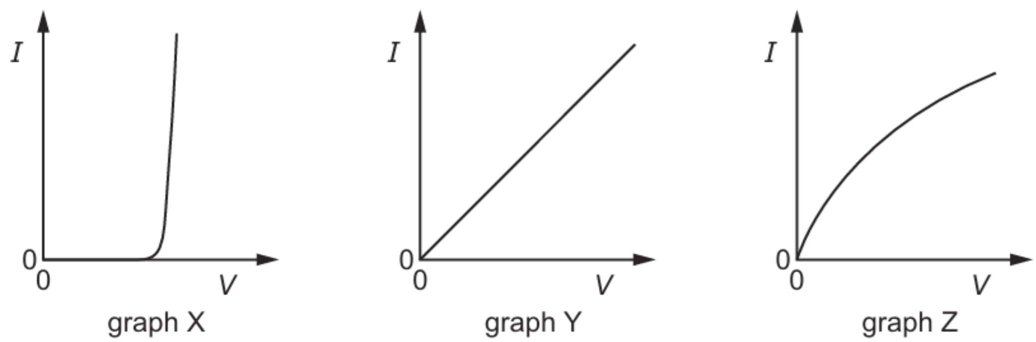
B The potential difference across the load resistor increases.

C The power dissipated in the internal resistance of the battery increases.

D The total power dissipated in the circuit increases.

232. 9702/12/O/N/21/Q35

The graphs show the variation with potential difference V of the current I for three circuit components.



The components are a metal wire at constant temperature, a semiconductor diode and a filament lamp.

Which row correctly identifies these graphs?

	metal wire at constant temperature	semiconductor diode	filament lamp
A	X	Z	Y
B	Y	X	Z
C	Y	Z	X
D	Z	X	Y

233. 9702/13/O/N/21/Q33

A lightning strike transfers 1×10^{20} electrons past a point in a time of $30\,\mu s$.

What is the average current during the lightning strike?

A 0.5 mA **B** 0.5 A **C** 500 A **D** 500 kA

234. 9702/13/O/N/21/Q35

A wire of uniform cross-section has resistance R.

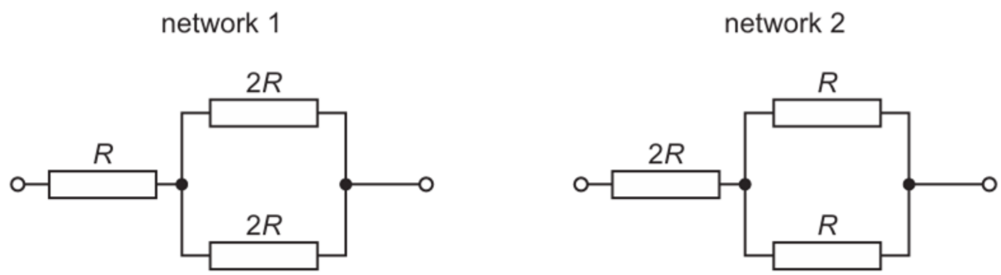
A second wire is made of the same material but is twice as long and has twice the diameter of the first wire.

What is the resistance of the second wire?

A $\frac{R}{8}$ **B** $\frac{R}{2}$ **C** R **D** $8R$

235. 9702/13/O/N/21/Q36

The diagram shows two resistor networks.



What is the ratio $\frac{\text{total resistance of network 1}}{\text{total resistance of network 2}}$?

A $\frac{4}{5}$ **B** $\frac{5}{4}$ **C** $\frac{1}{2}$ **D** $\frac{2}{1}$

236. 9702/12/F/M/22/Q29

For a current-carrying wire, the current can be calculated using the equation shown.

$$I = Anvq$$

What is the meaning of n?

A the number of charge carriers in the wire

B the number of charge carriers multiplied by the volume of the wire

C the number of charge carriers per unit length of the wire

D the number of charge carriers per unit volume of the wire

237. 9702/12/F/M/22/Q30

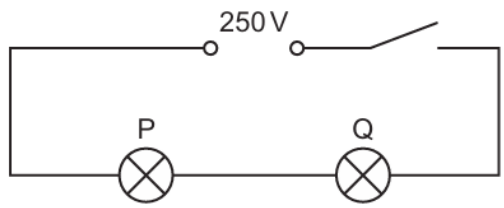
The number of free electrons passing a point in a wire in 24 hours is 6.0×10^{23}.

What is the average current in the wire?

A 6.3 pA **B** 1.1 A **C** 67 A **D** 4.0 kA

238. 9702/12/F/M/22/Q31

In the circuit shown, lamp P is rated 250 V, 50 W and lamp Q is rated 250 V, 200 W. The two lamps are connected in series to a 250 V power supply.



Assume that the resistance of each lamp remains constant.

Which statement most accurately describes what happens when the switch is closed?

A Lamp P emits four times as much power as lamp Q.

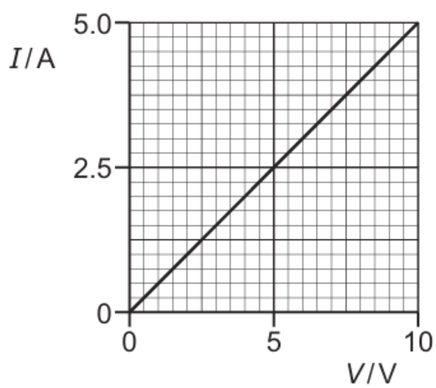
B Lamp P emits twice as much power as lamp Q.

C Lamp Q emits four times as much power as lamp P.

D Lamp Q emits twice as much power as lamp P.

239. 9702/12/F/M/22/Q32

A piece of wire has a length of 0.80 m and a diameter of 5.0×10^{-4} m. The I–V characteristic of the wire is shown.



What is the resistivity of the metal from which the wire is made?

A $1.2 \times 10^{-7}\,\Omega\,\text{m}$

B $1.6 \times 10^{-7}\,\Omega\,\text{m}$

C $4.9 \times 10^{-7}\,\Omega\,\text{m}$

D $2.0 \times 10^{-6}\,\Omega\,\text{m}$

240. 9702/11/M/J/22/Q31

A straight copper wire of diameter 0.42×10^{-3} m has a number density of free electrons of 8.5×10^{28} m^{-3}. In a given time interval, a charge of 0.15 C moves through the wire.

What is the average displacement of the free electrons along the wire in this time interval?

A 3.3×10^{-8} m **B** 2.0×10^{-5} m **C** 8.0×10^{-5} m **D** 2.5×10^{-4} m

241. 9702/11/M/J/22/Q32

What is the definition of the potential difference (p.d.) across a component?

A the electrical power supplied to the component

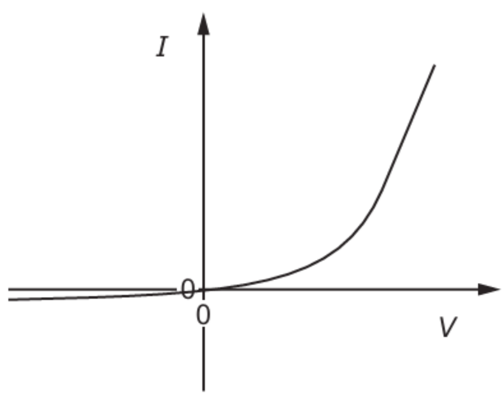
B the energy transferred to the component per unit charge

C the product of the current in the component and its resistance

D the voltage across the component

242. 9702/11/M/J/22/Q33

The graph shows the I–V characteristic for a semiconductor diode.



Which statement can be deduced from the graph?

A Above a certain positive potential difference the diode obeys Ohm's law.

B Current is directly proportional to potential difference when the current in the diode is in one direction.

C The diode has zero resistance when the current in the diode is in one direction.

D The resistance of the diode depends upon the potential difference across it.

243. 9702/11/M/J/22/Q34

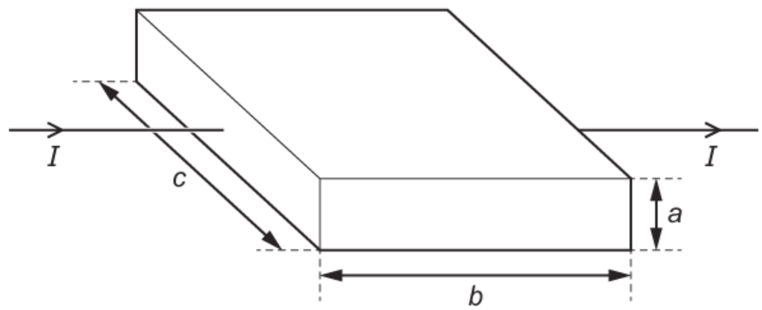
A wire has a resistance of 30 Ω. A second wire is made from the same material, has the same mass and is three times as long as the first wire.

What is the resistance of the second wire?

A 10 Ω **B** 30 Ω **C** 90 Ω **D** 270 Ω

244. 9702/12/M/J/22/Q31

The diagram shows a metal block.



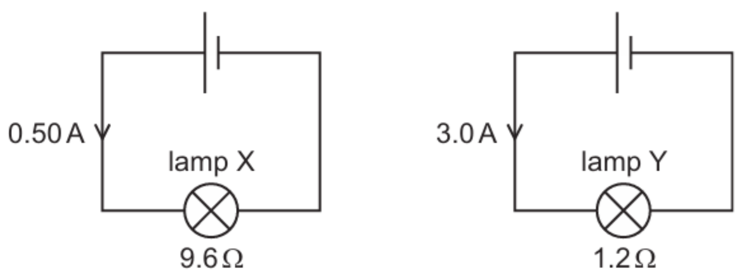
The block has sides of length *a*, *b* and *c* as shown, and its volume is *V*. Each charge carrier has a charge $-q$ and the number density of the charge carriers in the metal is *n*. It takes each charge carrier an average time of *t* to pass through the block.

What is an expression for the current I?

A $I = nqabc$ **B** $I = \dfrac{nqV}{t}$ **C** $I = \dfrac{nqbc}{t}$ **D** $I = nqaV$

245. 9702/12/M/J/22/Q32

The circuit diagrams show two lamps X and Y each connected to a cell. The current in lamp X is 0.50 A and its resistance is 9.6 Ω. The current in lamp Y is 3.0 A and its resistance is 1.2 Ω.



What is the ratio $\dfrac{\text{power in lamp X}}{\text{power in lamp Y}}$?

A 0.22 **B** 0.75 **C** 1.3 **D** 4.5

246. 9702/12/M/J/22/Q33

The intensity of light incident on a light-dependent resistor (LDR) is increased. The temperature of a thermistor is increased. In each case, the current in the component is maintained at a constant value.

What happens to the potential difference across each component?

	LDR	thermistor
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

247. 9702/12/M/J/22/Q34

An iron wire has length 8.0 m and diameter 0.50 mm. The wire has resistance R.

A second iron wire has length 2.0 m and diameter 1.0 mm.

What is the resistance of the second wire?

A $\frac{R}{16}$ **B** $\frac{R}{8}$ **C** $\frac{R}{2}$ **D** R

248. 9702/13/M/J/22/Q32

Two wires, X and Y, are made from the same metal.

The diameter of wire Y is twice that of wire X.

Wire X, wire Y and a battery are connected in series.

What is the ratio $\dfrac{\text{average drift speed of free electrons in wire X}}{\text{average drift speed of free electrons in wire Y}}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** $\frac{2}{1}$ **D** $\frac{4}{1}$

249. 9702/13/M/J/22/Q33

A resistor has resistance R. When the potential difference (p.d.) across the resistor is V, the current in the resistor is I. The power dissipated in the resistor is P. Work W is done when charge Q flows through the resistor.

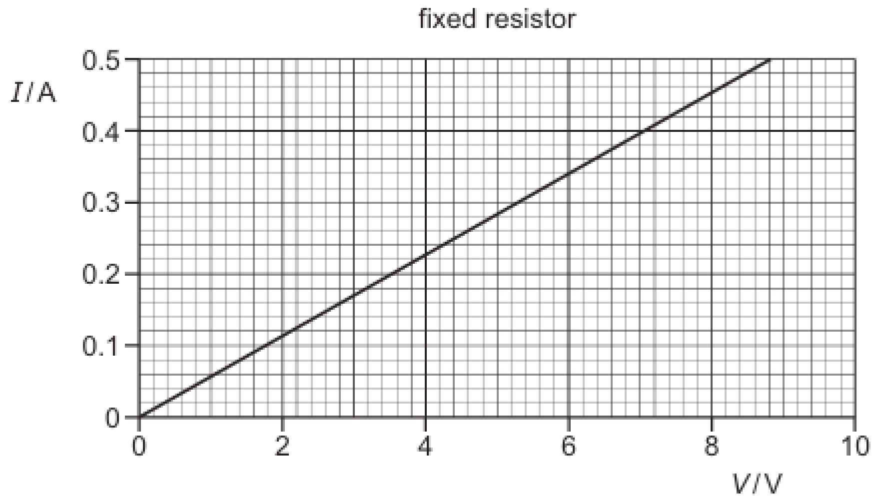
What is **not** a valid relationship between these variables?

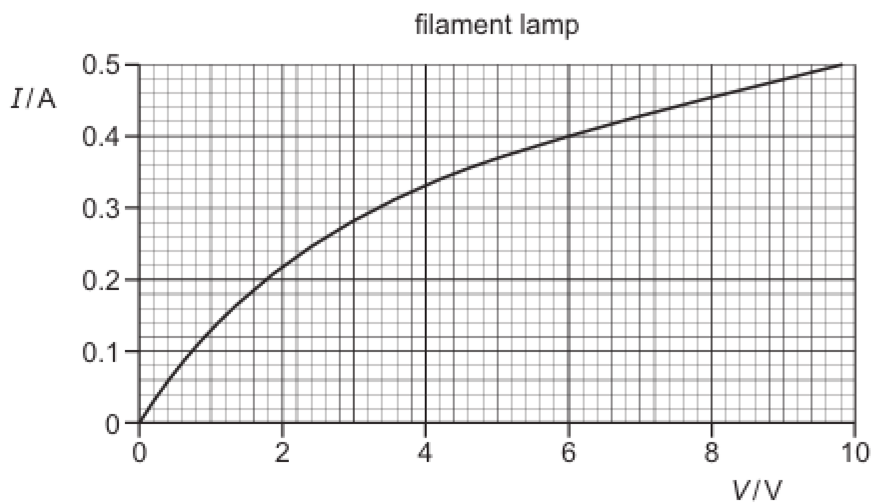
A $I = \frac{P}{V}$ **B** $Q = \frac{W}{V}$ **C** $R = \frac{P}{I^2}$ **D** $R = \frac{V}{P}$

250. 9702/13/M/J/22/Q34

A fixed resistor and a filament lamp are connected in series to a power supply.

The I–V characteristics for the two components are shown.





The current in the fixed resistor is 0.34 A.

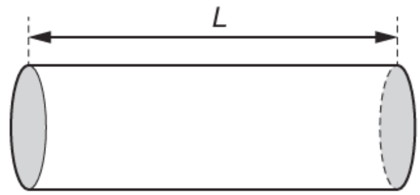
What is the resistance of the filament lamp?

A 0.081 Ω **B** 12 Ω **C** 15 Ω **D** 18 Ω

251. 9702/13/M/J/22/Q35

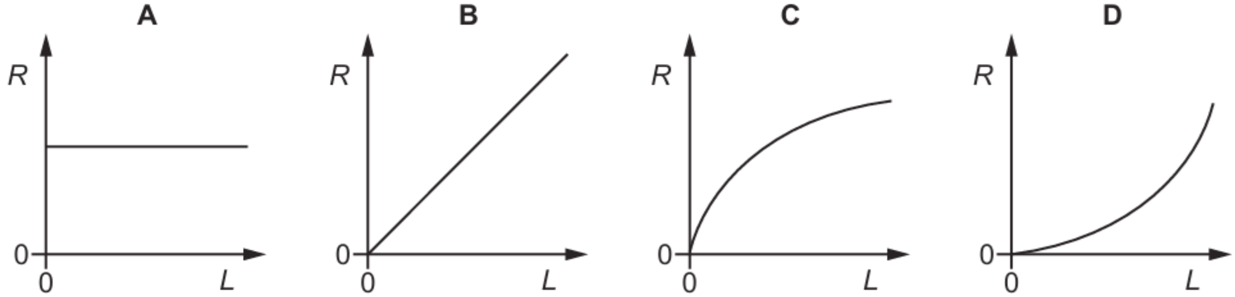
A piece of conducting putty (modelling clay) of constant resistivity is formed into a cylindrical shape.

The resistance R between its flat ends (shaded) is measured.



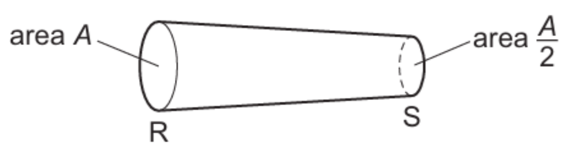
The same volume of putty is re-formed into cylinders of different lengths L, and the resistance R between the flat ends is measured for each value of L.

Which graph best shows the variation of R with L?



252. 9702/11/O/N/22/Q31

A length of wire RS has a circular cross-section.



At end R of the wire, the cross-sectional area is A.

At end S of the wire, the cross-sectional area is $\frac{A}{2}$.

Charge Q takes time t to pass through end R of the wire. There is a constant electric current in the wire.

How much charge will pass through end S in a time interval of $\frac{t}{4}$?

A $\frac{Q}{8}$ **B** $\frac{Q}{4}$ **C** $\frac{Q}{2}$ **D** Q

253. 9702/11/O/N/22/Q32

A power supply is connected to a component by connecting wires of total resistance 4.9 Ω.

The power supply has an output power of 3.6 W and a terminal potential difference of 12 V.

How much thermal energy is dissipated in the connecting wires in a time of 1.0 hour?

A 0.44 J **B** 29 J **C** 1.6 kJ **D** 11 kJ

254. 9702/11/O/N/22/Q33

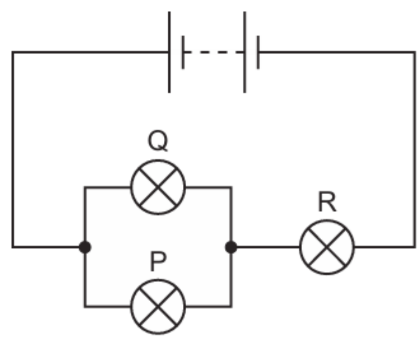
A copper wire is to be replaced by an aluminium alloy wire of the same length and resistance. Copper has half the resistivity of the alloy.

What is the ratio $\dfrac{\text{diameter of alloy wire}}{\text{diameter of copper wire}}$?

A $\sqrt{2}$ **B** 2 **C** $2\sqrt{2}$ **D** 4

255. 9702/11/O/N/22/Q34

Three identical filament lamps, P, Q and R, are connected to a battery of negligible internal resistance, as shown.



The filament wire in lamp Q breaks so that it no longer conducts.

What are the changes in the brightness of lamps P and R?

	lamp P	lamp R
A	brighter	brighter
B	brighter	dimmer
C	dimmer	brighter
D	dimmer	dimmer

256. 9702/12/O/N/22/Q31

A nichrome wire has a resistance of 15 Ω and a diameter of 3.0 mm. The number density of the free electrons in nichrome is $9.0 \times 10^{28}\,m^{-3}$.

A potential difference (p.d.) of 6.0 V is applied between the ends of the wire.

What is the average drift speed of the free electrons in the wire?

A $9.8 \times 10^{-7}\,m\,s^{-1}$

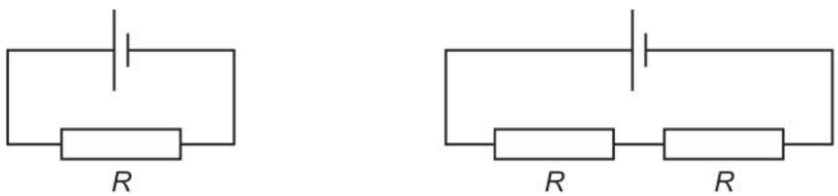
B $3.9 \times 10^{-6}\,m\,s^{-1}$

C $6.1 \times 10^{-6}\,m\,s^{-1}$

D $2.5 \times 10^{-5}\,m\,s^{-1}$

257. 9702/12/O/N/22/Q32

The diagrams show two different circuits.



The cells in each circuit have the same electromotive force (e.m.f.) and negligible internal resistance. The three resistors each have the same resistance *R*.

In the circuit on the left, the power dissipated in the resistor is *P*.

What is the total power dissipated in the circuit on the right?

A $\frac{P}{4}$ **B** $\frac{P}{2}$ **C** P **D** $2P$

258. 9702/12/O/N/22/Q33

The potential difference (p.d.) across a filament lamp is increased.

Which statement is correct?

A The resistance of the lamp decreases because the temperature decreases.

B The resistance of the lamp decreases because the temperature increases.

C The resistance of the lamp increases because the temperature decreases.

D The resistance of the lamp increases because the temperature increases.

259. 9702/12/O/N/22/Q34

A metal wire has resistance *R*.

The wire is stretched so that its diameter decreases to 94.0% of the original diameter.

The volume of the wire is unchanged.

What is the resistance of the stretched wire?

A 1.06*R* **B** 1.13*R* **C** 1.20*R* **D** 1.28*R*

260. 9702/13/O/N/22/Q30

There is an electric current in a copper wire.

Which statement describing the average drift speed of the charge carriers in the wire is correct?

A It is nearly $3 \times 10^8 \, m \, s^{-1}$.

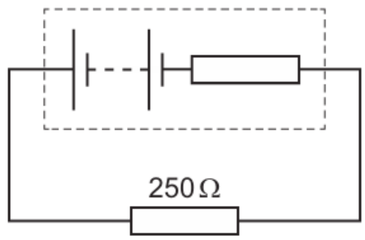
B It is proportional to the cross-sectional area of the wire.

C It is proportional to the length of the wire.

D It is proportional to the magnitude of the current.

261. 9702/13/O/N/22/Q31

A battery with a constant internal resistance is connected to a resistor of resistance $250\,\Omega$, as shown.



The current in the resistor is 40 mA for a time of 60 s. During this time 6.0 J of energy is dissipated by the internal resistance.

What is the energy supplied to the external resistor during the 60 s and the electromotive force (e.m.f.) of the battery?

	energy / J	e.m.f. / V
A	30	2.5
B	30	7.5
C	24	10.0
D	24	12.5

262. 9702/13/O/N/22/Q33

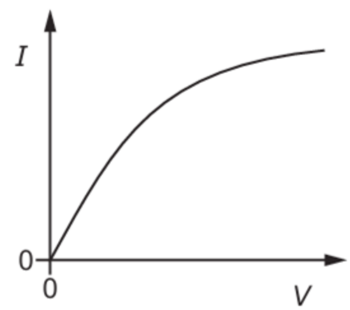
Two wires, P and Q, have the same resistance. Wire Q is made of material that has twice the resistivity of the material used to make wire P. The diameter of wire Q is twice the diameter of wire P.

What is the ratio $\dfrac{\text{length of wire P}}{\text{length of wire Q}}$?

A $\dfrac{1}{8}$ **B** $\dfrac{1}{4}$ **C** $\dfrac{1}{2}$ **D** $\dfrac{2}{1}$

263. 9702/13/O/N/22/Q32

Which component has the I–V graph shown?



A filament lamp

B metallic conductor at constant temperature

C resistor of fixed resistance

D semiconductor diode

264. 9702/12/F/M/23/Q30

A wire carries a current of 0.10 μA. The potential difference across the wire is 10 mV.

How much energy is dissipated by the wire in a time of 10 s?

A 1.0 pJ **B** 10 pJ **C** 1.0 nJ **D** 10 nJ

265. 9702/12/F/M/23/Q31

What is the definition of the potential difference across an electrical component?

A energy transferred per unit charge

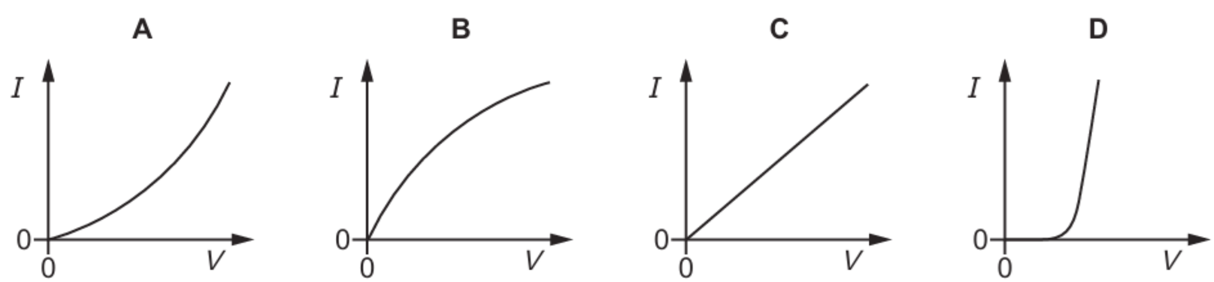
B energy transferred per unit current

C energy transferred per unit resistance

D energy transferred per unit time

266. 9702/12/F/M/23/Q32

Which graph represents the way the current I through a filament lamp varies with the potential difference V across it?



267. 9702/12/F/M/23/Q33

The table shows the properties of two different wires, P and Q.

	length	resistance	resistivity of material
wire P	l	R	ρ
wire Q	$2l$	$\frac{1}{4}R$	$\frac{1}{3}\rho$

Wire P has a cross-section of diameter d.

What is the diameter of the cross-section of wire Q?

A $0.41d$ **B** $1.6d$ **C** $2.7d$ **D** $7.1d$

268. 9702/11/M/J/23/Q30

The electric current in a metal wire is 4.0 mA.

How many electrons pass a fixed point in the wire in a time of 10 hours?

A 2.5×10^{17} **B** 2.5×10^{20} **C** 9.0×10^{20} **D** 9.0×10^{23}

269. 9702/11/M/J/23/Q31

What is the definition of the potential difference across an electrical component?

A the charge per unit time passing through the component

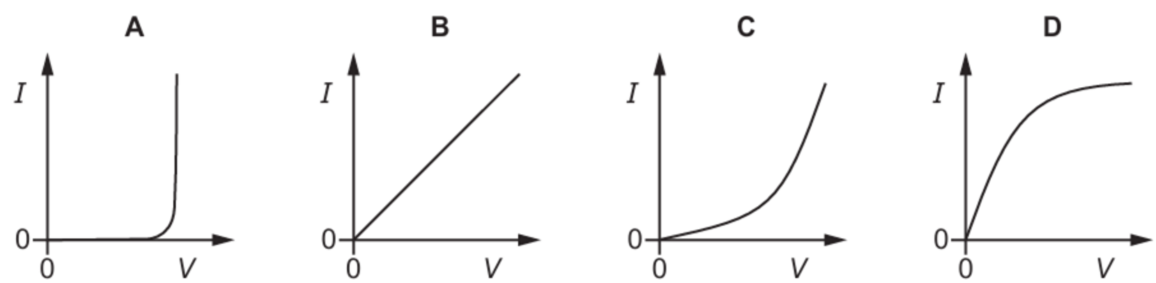
B the energy transferred per unit charge

C the force per unit charge

D the resistance per unit current

270. 9702/11/M/J/23/Q32

Which graph shows the I–V characteristic of a filament lamp?



271. 9702/11/M/J/23/Q33

A metal wire has a length of 2.50 m and a cross-sectional area of 4.50×10^{-6} m^2. The resistivity of the metal is 3.50×10^{-7} Ω m.

The wire is stretched so that its length increases to 2.65 m. The wire remains cylindrical and the **volume** of the wire remains constant.

What is the change in the resistance of the wire?

A 0.012 Ω **B** 0.024 Ω **C** 0.19 Ω **D** 0.22 Ω

272. 9702/12/M/J/23/Q30

Which charge can be carried by a charge carrier?

A 1.1×10^{-19} C

B 4.0×10^{-19} C

C 4.8×10^{-19} C

D 9.1×10^{-19} C

273. 9702/12/M/J/23/Q31

A resistor of resistance *R* is connected across a cell of electromotive force (e.m.f.) *E* and negligible internal resistance.

Which single change to the circuit would lead to the largest increase in the power dissipated in the resistor?

A doubling the value of *E*

B doubling the value of *R*

C halving the value of *E*

D halving the value of *R*

274. 9702/12/M/J/23/Q32

An electrical cable is made up of one thick strand of copper wire that is surrounded by eight thin strands of copper wire. All nine strands of wire are connected in parallel with each other.

A cross-section of the cable is shown.

Each thin strand of wire has cross-sectional area *A* and length *L*.

The thick strand of wire has cross-sectional area 4*A* and length *L*.

The cable has total resistance *R*.

Which expression gives the resistivity of copper?

A $\dfrac{4A}{33RL}$ **B** $\dfrac{12A}{RL}$ **C** $\dfrac{4AR}{L}$ **D** $\dfrac{12AR}{L}$

275. 9702/12/M/J/23/Q33

A car has sensors for detecting the light intensity and temperature of its environment.

The sensors make use of a light-dependent resistor (LDR) and a thermistor.

The car moves from a warm and dark environment into a cold and bright environment.

What are the changes to the resistances of the LDR and thermistor?

	resistance of LDR	resistance of thermistor
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

276. 9702/13/M/J/23/Q30

A metal wire is connected between the terminals of a cell so that there is a current in the wire.

Which statement is correct?

A Negatively charged electrons in the wire move from the negative terminal to the positive terminal.

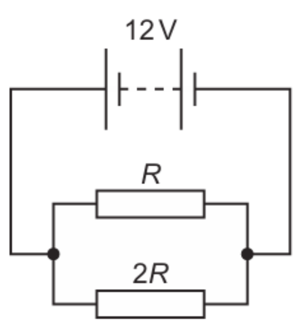
B Negatively charged nuclei in the wire move from the negative terminal to the positive terminal.

C Positively charged electrons in the wire move from the positive terminal to the negative terminal.

D Positively charged nuclei in the wire move from the positive terminal to the negative terminal.

277. 9702/13/M/J/23/Q31

Two resistors of resistances *R* and 2*R* are connected in parallel with a battery of electromotive force (e.m.f.) 12 V and negligible internal resistance.



The total power dissipated by the two resistors is 36 W.

What is the value of *R*?

A 0.50 Ω **B** 2.7 Ω **C** 4.0 Ω **D** 6.0 Ω

278. 9702/13/M/J/23/Q32

A wire has a length of 3.0 m and is made of metal of resistivity $4.9 \times 10^{-7}\,\Omega\,m$.

A potential difference (p.d.) of 12 V is applied across the wire so that it has a current of 1.4 A.

What is the cross-sectional area of the wire?

A $1.2 \times 10^{-7}\,m^2$

B $1.7 \times 10^{-7}\,m^2$

C $1.1 \times 10^{-6}\,m^2$

D $1.3 \times 10^{-5}\,m^2$

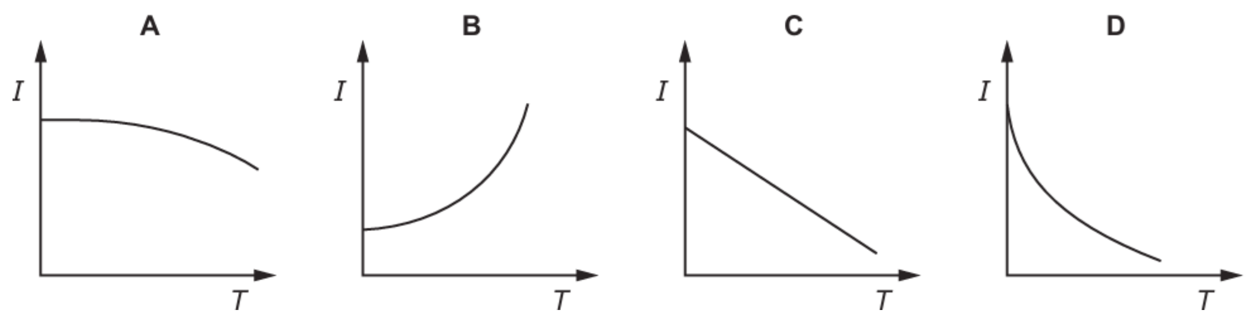
279. 9702/13/M/J/23/Q33

A cell of negligible internal resistance is connected in series with a thermistor, a fixed resistor and an ammeter.

The thermistor is placed in a beaker of water and the temperature of the water is slowly increased.

A graph of current I against the temperature T of the thermistor is plotted.

Which graph could show the variation of I with T?



280. 9702/11/O/N/23/Q30

The current I in a conductor is given by the equation shown.

$$I = Anvq$$

What does the letter n represent in this equation?

A charge carried per charge carrier

B number of charge carriers per unit area

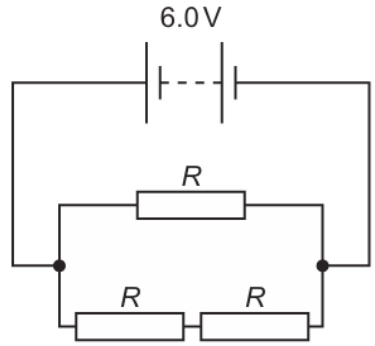
C number of charge carriers per unit volume

D total mass of charge carriers per unit volume

281. 9702/11/O/N/23/Q31

In the circuit shown, the battery has an electromotive force (e.m.f.) of 6.0 V and negligible internal resistance.

The three resistors each have resistance R.



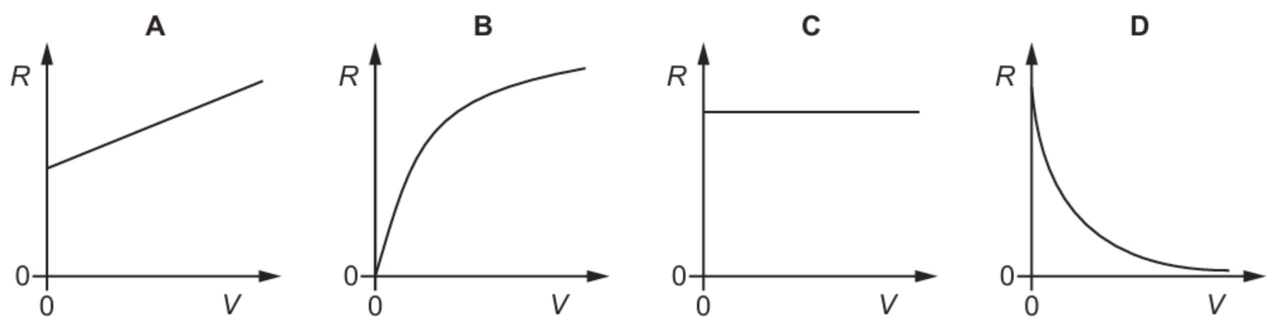
The total power dissipated in the resistor network is 24 W.

What is the value of R?

A 0.50 Ω **B** 1.0 Ω **C** 1.5 Ω **D** 2.3 Ω

282. 9702/11/O/N/23/Q32

Which graph could show how the resistance R of a filament lamp varies with the applied potential difference (p.d.) V, as V is increased to the normal operating p.d.?



283. 9702/11/O/N/23/Q33

A piece of conducting putty is in the shape of a cylinder of length 60 mm and diameter 20 mm. The resistance between the ends of the cylinder is 20 Ω.

What is the resistivity of the putty?

A 0.033 Ωm **B** 0.10 Ωm **C** 0.42 Ωm **D** 5.2 Ωm

284. 9702/12/O/N/23/Q30

A fine mist of oil droplets is sprayed into air. As the oil droplets leave the nozzle of the spraying device they can become electrically charged.

What is **not** a possible value for the charge on an oil droplet?

A zero

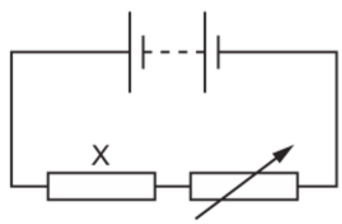
B 1.0×10^{-19} C

C 4.8×10^{-19} C

D 8.0×10^{-19} C

285. 9702/12/O/N/23/Q31

In the circuit shown, a fixed resistor X is connected in series with a battery and a variable resistor.



The power dissipated in resistor X is 7.2 W when a current of 3.0 A passes through it.

The variable resistor is adjusted so that the power dissipated in X increases by 50%.

What is the new current in the circuit?

A 2.4 A **B** 3.7 A **C** 4.5 A **D** 14 A

286. 9702/12/O/N/23/Q32

The potential difference across a metal wire is kept constant. The length l and the diameter d of the wire are both varied. The type of metal is kept the same.

How is the current in the wire related to l and d?

A It is directly proportional to l and inversely proportional to d.

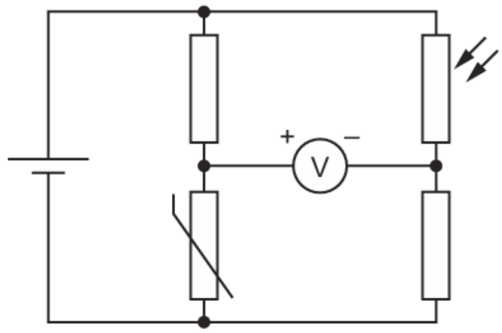
B It is directly proportional to l and inversely proportional to d^2.

C It is inversely proportional to l and directly proportional to d.

D It is inversely proportional to l and directly proportional to d^2.

287. 9702/12/O/N/23/Q33

A student sets up a circuit. The circuit diagram shows how the positive and negative terminals of a voltmeter are connected to the circuit. The voltmeter has an initial reading that is positive.



Which changes, if any, in temperature and light intensity would cause the voltmeter reading to **decrease**?

	temperature	light intensity
A	increase	increase
B	no change	decrease
C	decrease	no change
D	decrease	decrease

288. 9702/13/O/N/23/Q30

What could **not** be used to create an electric current?

A alpha-particles

B beta-particles

C neutrons

D protons

289. 9702/13/O/N/23/Q31

What is the definition of the potential difference (p.d.) across a component?

A the energy transferred per unit charge

B the energy transferred per unit current

C the power transferred per unit charge

D the power transferred per unit current

290. 9702/13/O/N/23/Q32

The resistance of a filament lamp increases as the current in it increases.

What is the reason for this?

A The charge of each charge carrier increases.

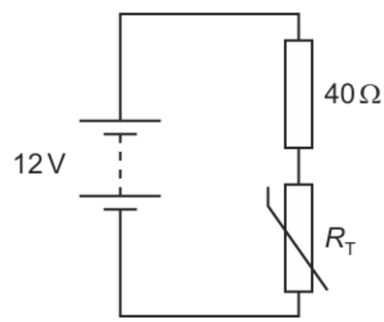
B The potential difference across the filament decreases.

C The power dissipated by the filament decreases.

D The temperature of the filament increases.

291. 9702/13/O/N/23/Q33

A battery of electromotive force (e.m.f.) 12 V and negligible internal resistance is connected to a fixed resistor of resistance 40 Ω and a thermistor of resistance R_T, as shown.



Initially, the temperature of the thermistor is 15 °C and the current in the circuit is 0.10 A.

The temperature of the thermistor then changes, which causes the current to increase to 0.12 A.

How does the temperature of the thermistor change and what is R_T at the new temperature?

	temperature of thermistor	R_T at new temperature / Ω
A	increases	60
B	decreases	60
C	increases	100
D	decreases	100